DEPARTMENT OF INFORMATION TECHNOLOGY

ENGINEERING LAB MANUAL

CS23432 - SoftwareConstruction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

Thandalam, Chennai-602015

Name: **V.CHARAN SUNDAR**

Register No: 241001036

Year / Branch / Section: II/B.Tech/IT - A

Semester: IV

Academic Year: 2025-2026

INDEX

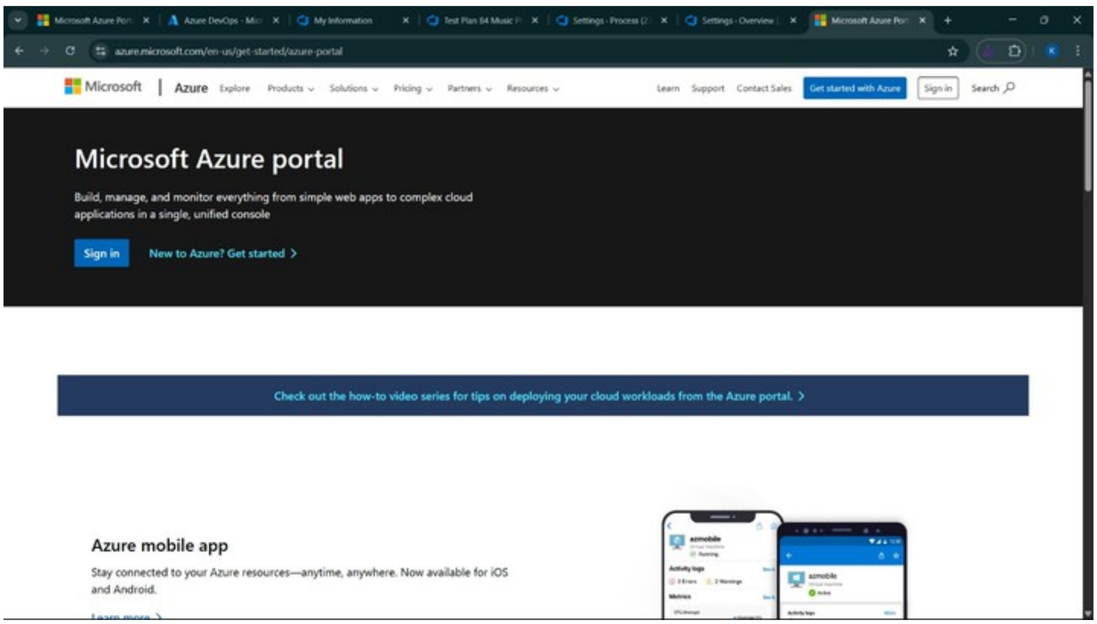
EXP NO: 1	AZUREDEVOPS ENVIRONMENT SETUP

Aim:

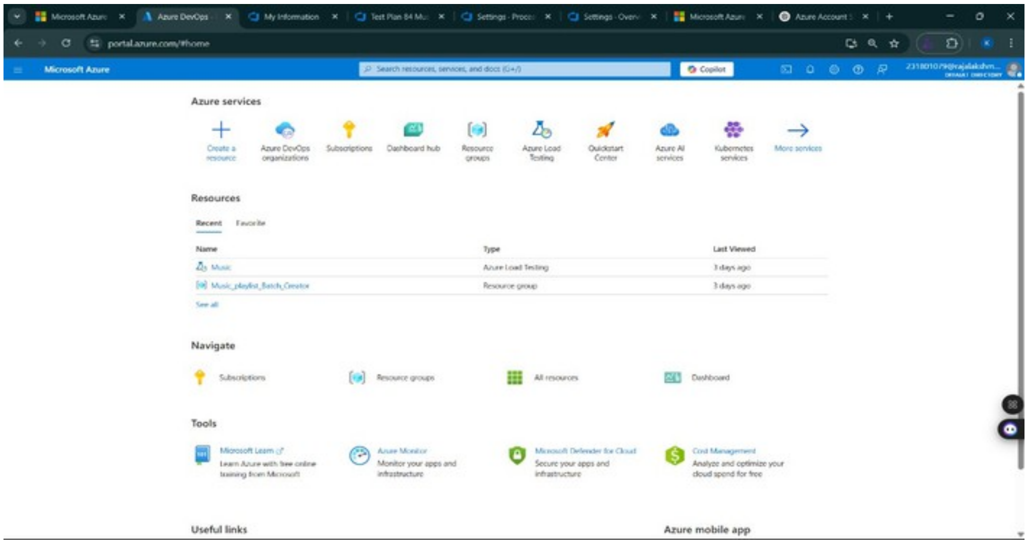
To set up and access the Azure DevOps environment by creating an organization through the Azure portal for the Hospital Management System project.

INSTALLATION

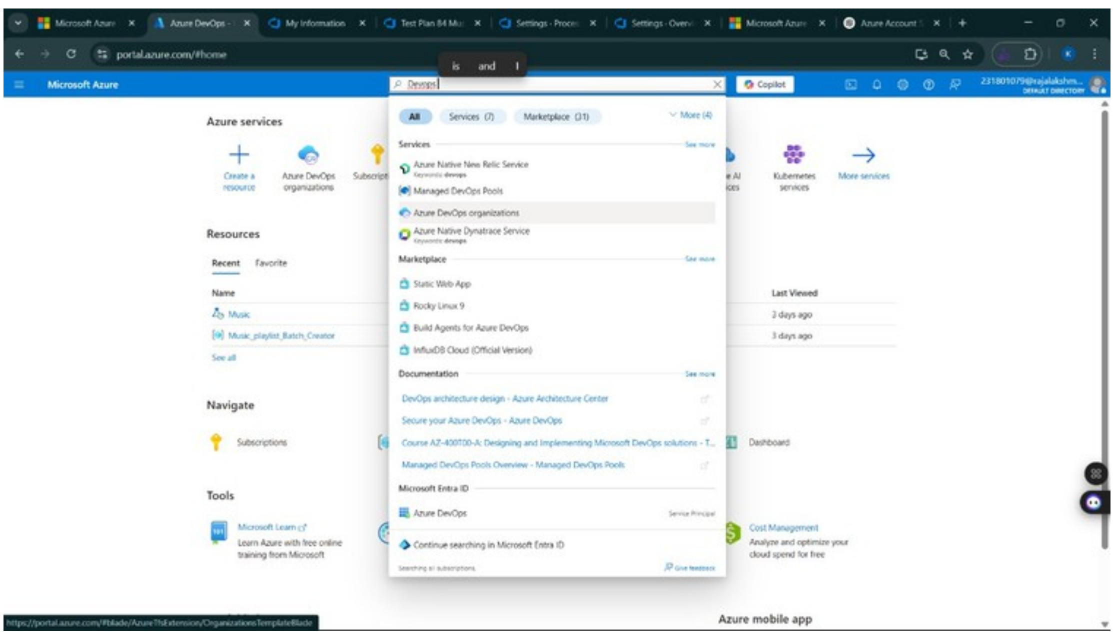
1.Open your web browser and go to the Azure website: https://azure.microsoft.com/en-us/get-started/azure-portal . Sign in using your Microsoft account credentials.
If you don't have a Microsoft account, you can create one here: https://signup.live.com/?lic=1



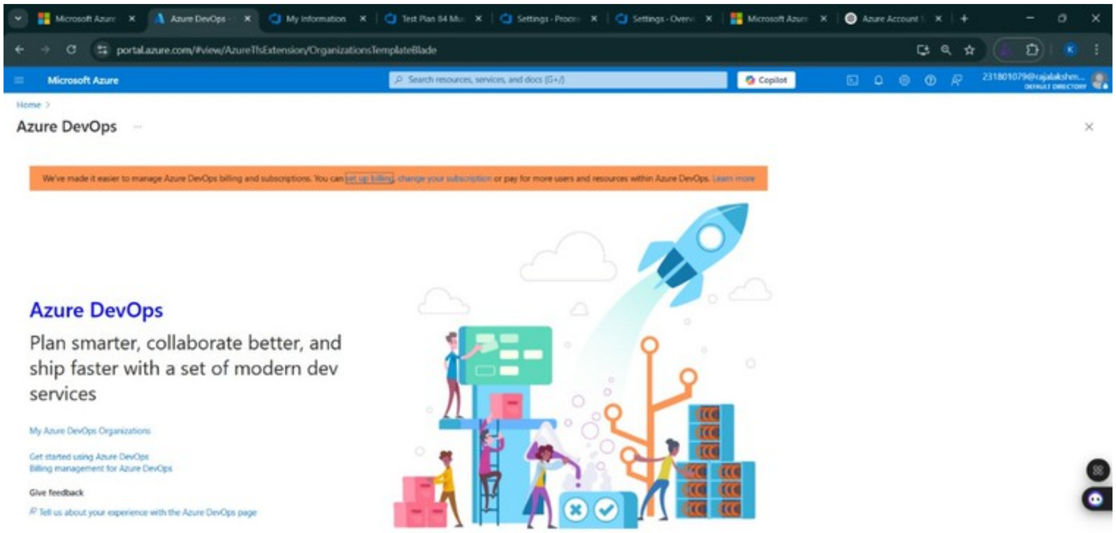
2.Azure home page



3.Open DevOps environment in the Azure platform by typing ***Azure DevOps Organizations*** in the search bar.



4.Click on the ***My Azure DevOps Organization*** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



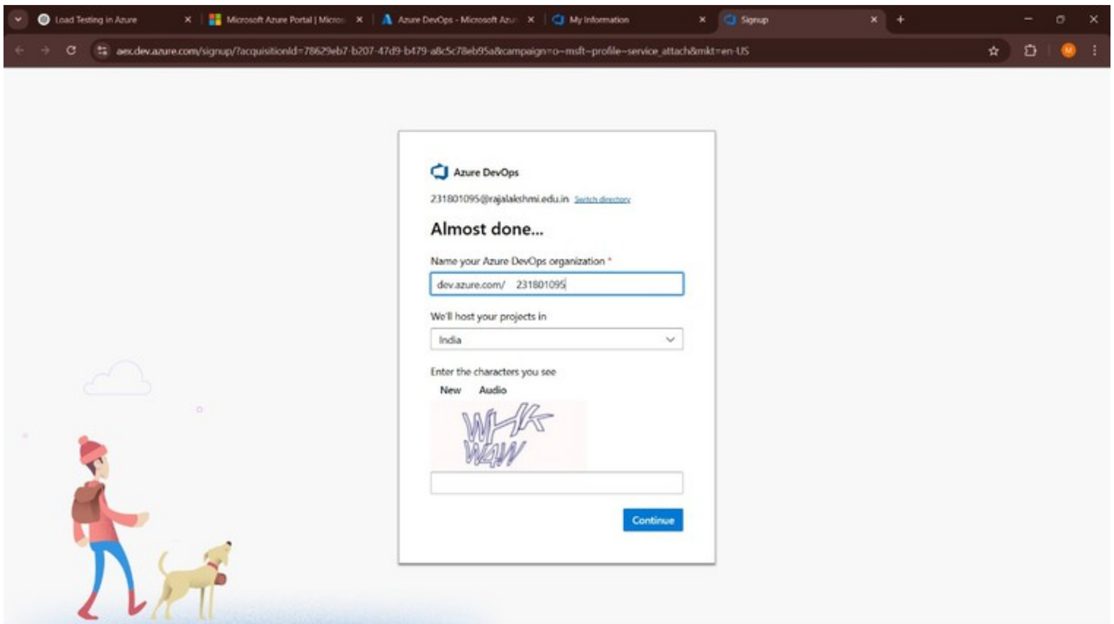
Result:

Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2	**AZUREDEVOPS PROJECT SETUP AND USER STORY MANAGEMENT**

Aim:
To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



2.Create the First Project in Your Organization

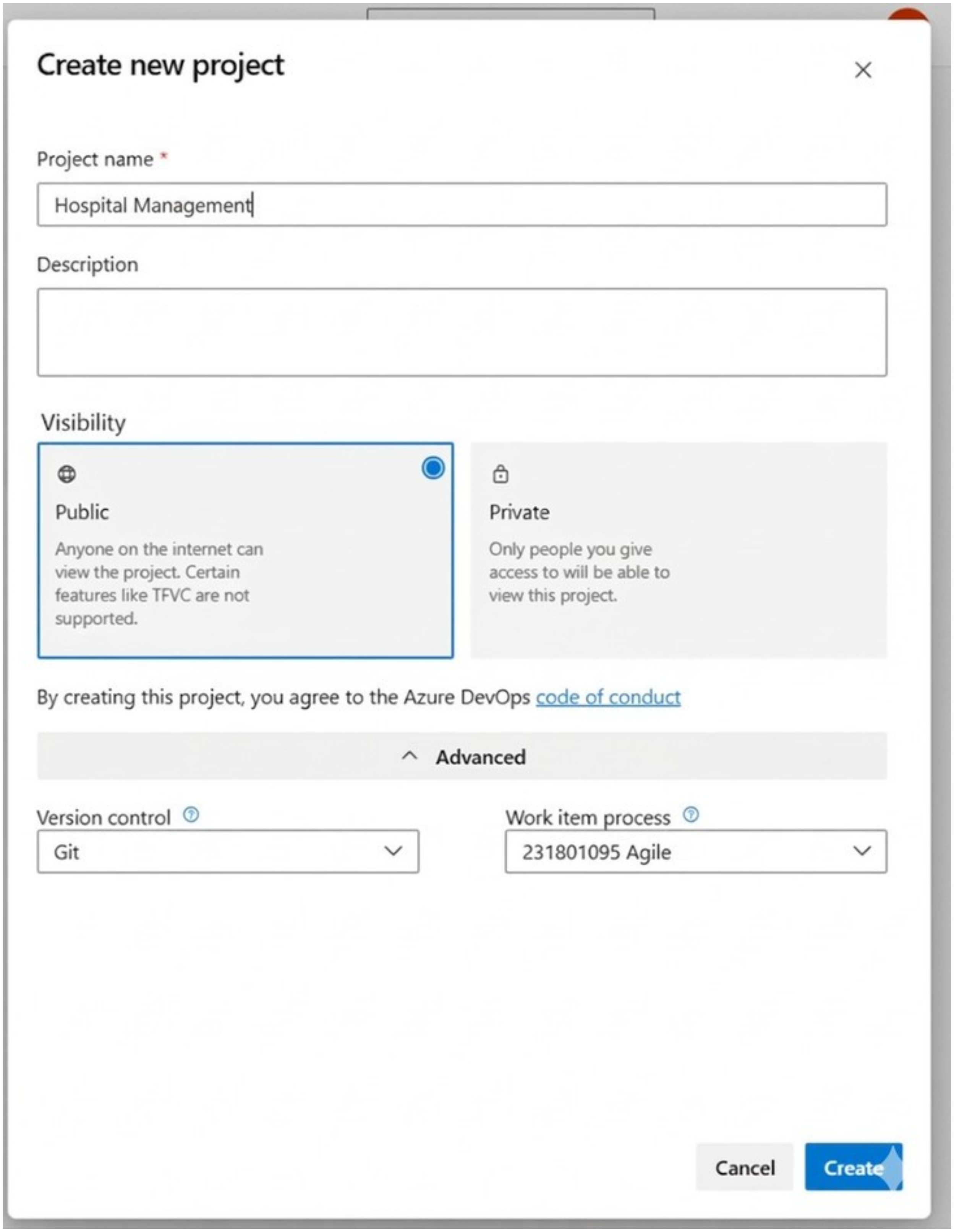
a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization's **Home page**, click on the **New Project** button.

c. Enter the project name, description, and visibility options:

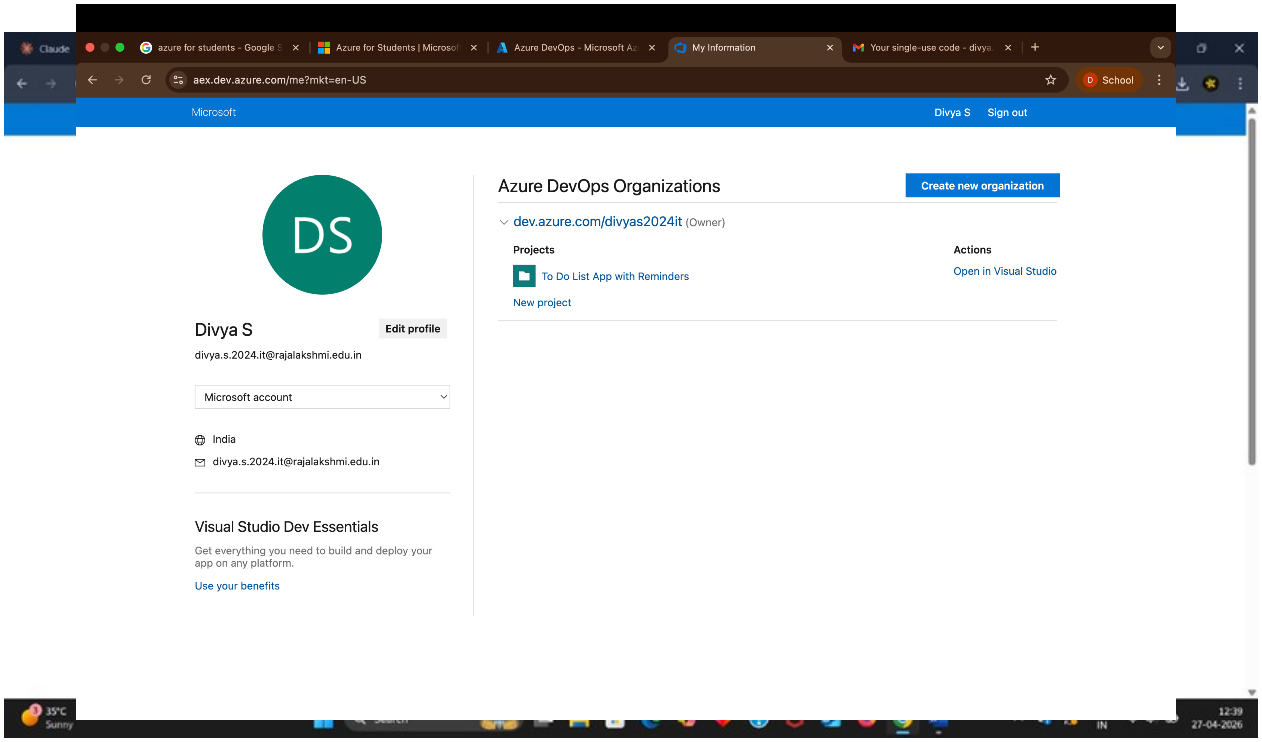
Name: Choose a name for the project (e.g., Hospital Management System (HMS)). ***Description***: Optionally, add a description to provide more context about the project. ***Visibility***: Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).

d. Once you've filled out the details, click **Create** to set up your first project.



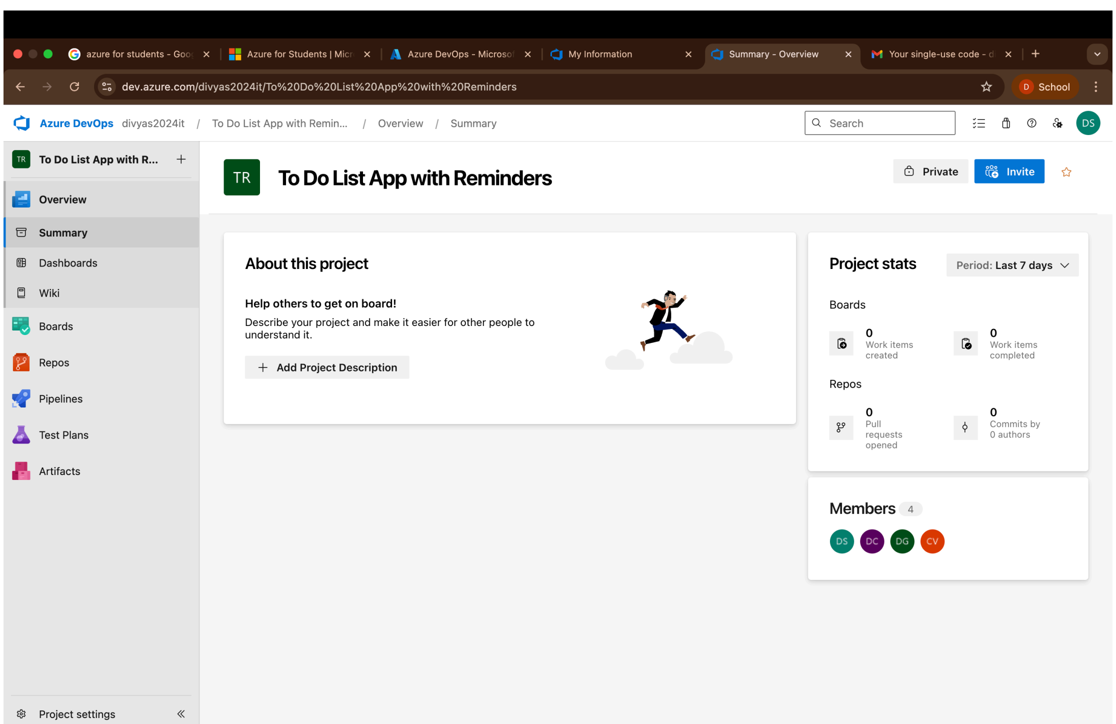
Create new project
Project name *
Hospital Management
Description
Visibility
Public
Anyone on the internet can view the project. Certain features like TFVC are not supported.
Private
Only people you give access to will be able to view this project.
By creating this project, you agree to the Azure DevOps code of conduct
Advanced
Version control
Git
Work item process
231801095 Agile
Cancel
Create

3.Once logged in, ensure you are in the correct organization. If you're part of multiple



organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.

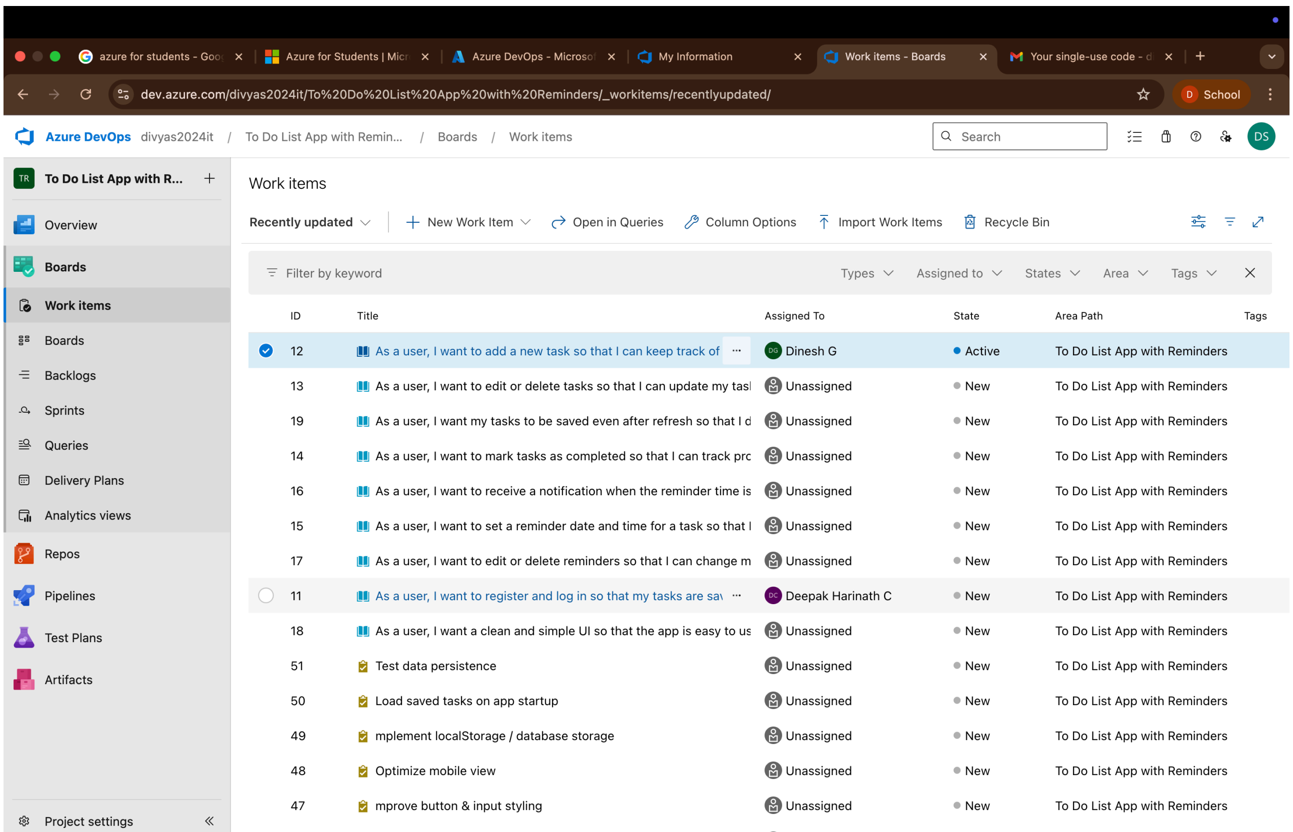
4.Project dashboard

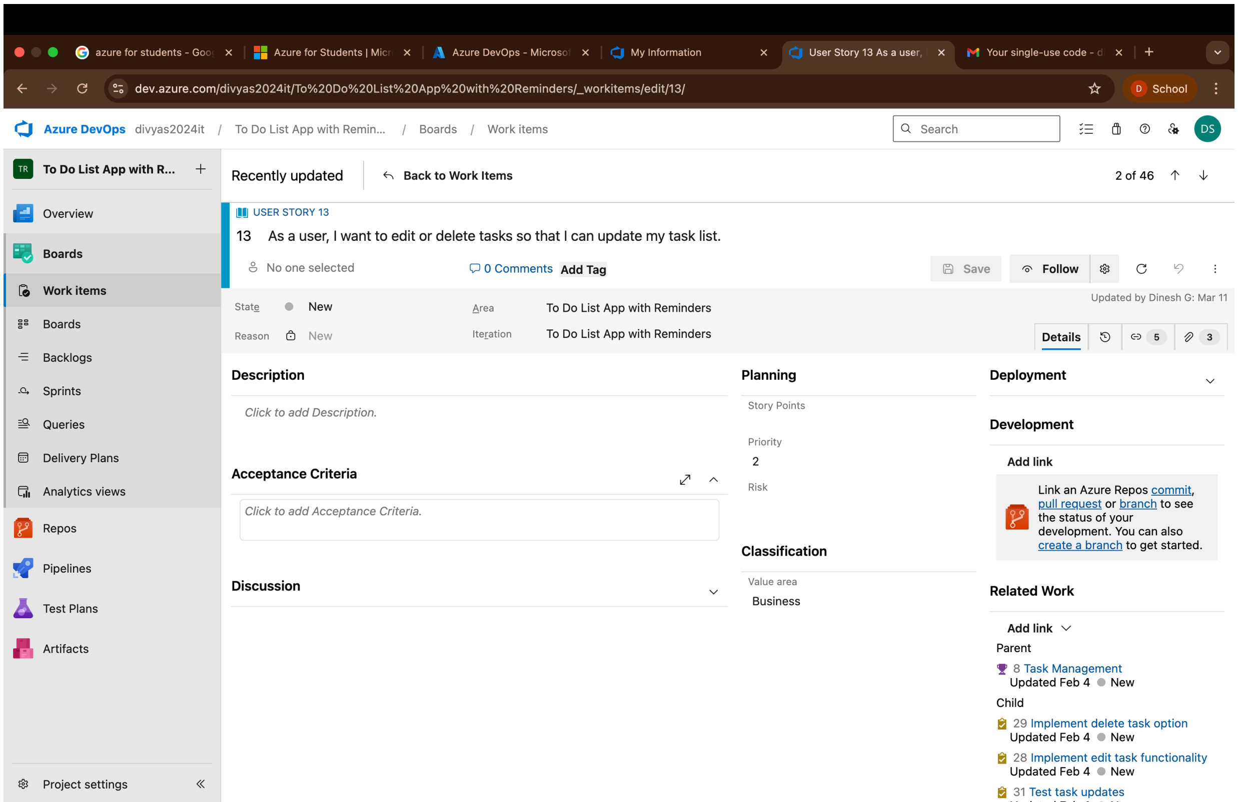


5.To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a **+** button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.





RESULT

Successfully created an Azure DevOps project with user story management and agile workflow

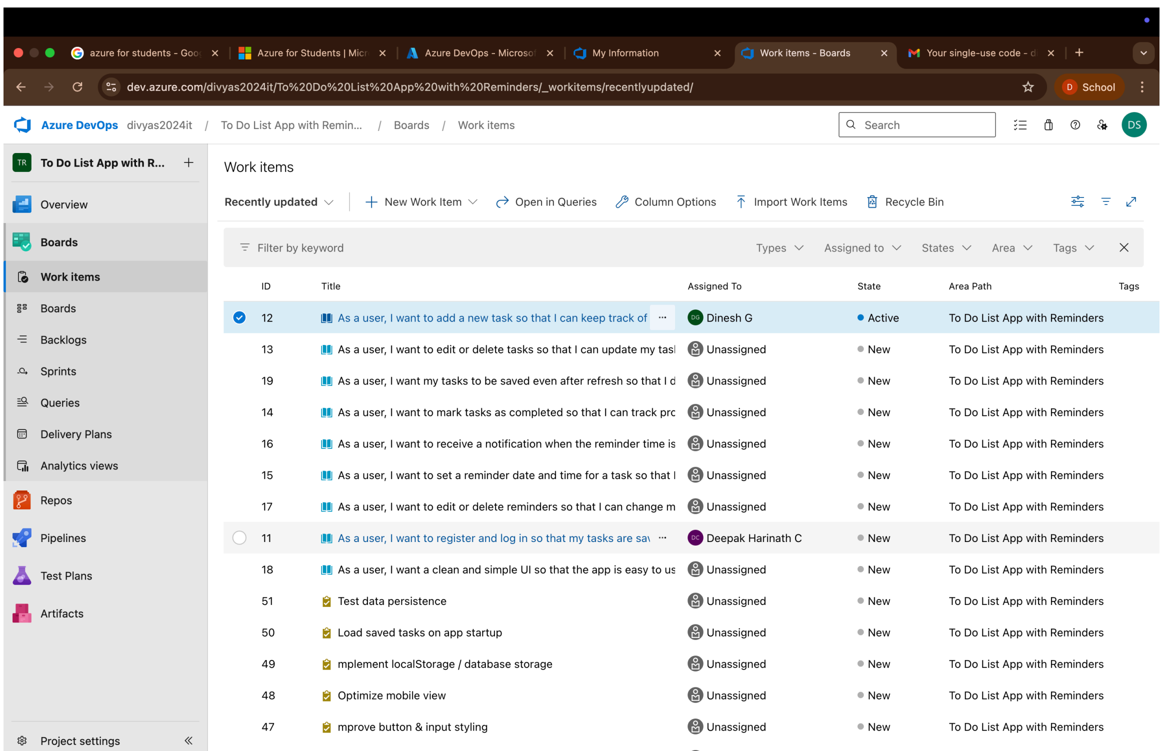
EX NO.3	**SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING**

Aim:

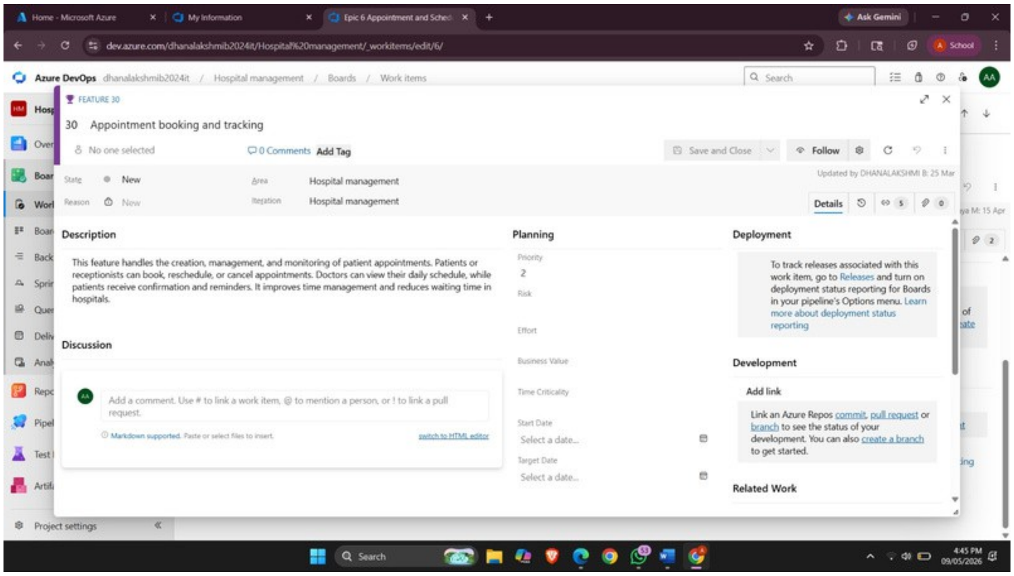
To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task

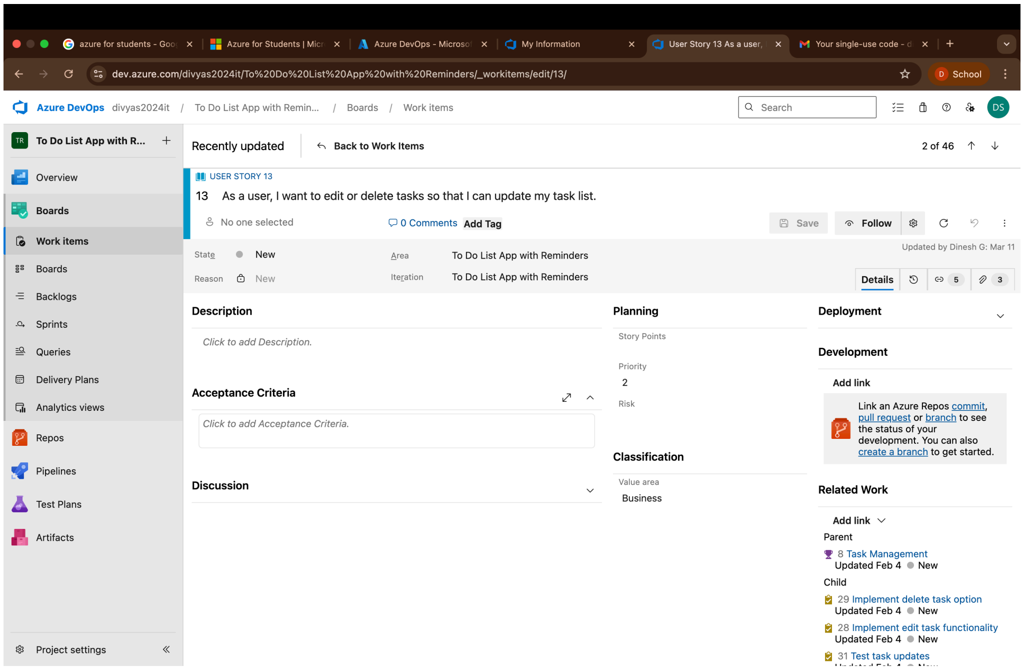
1.Fill in Epics



2. Fill in Features



3. Fill in User Story Details



Result:

Thus, the creation of epics, features, user story and task has been created successfully.

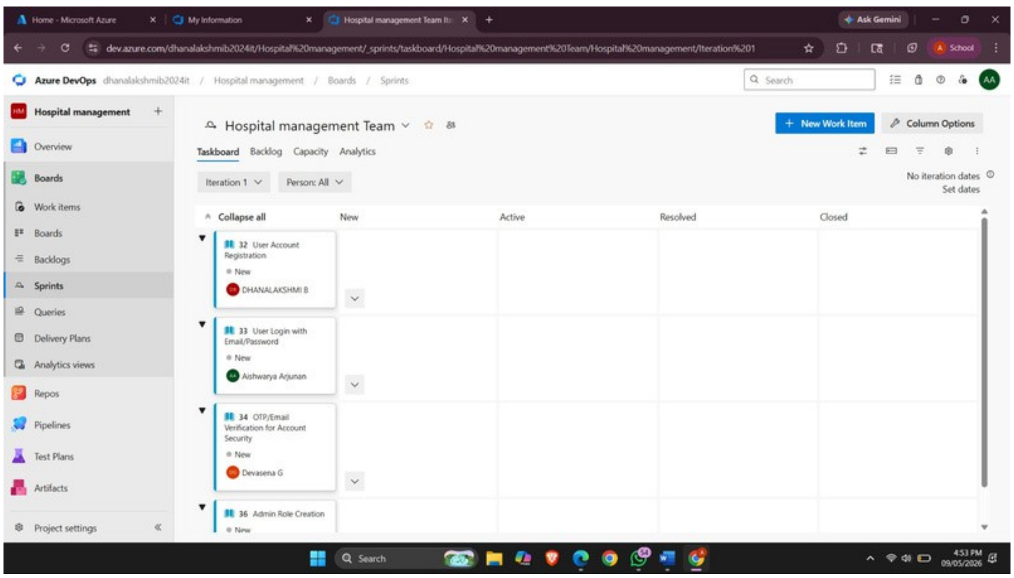
EXP NO: 4	SPRINT PLANNING

Aim:

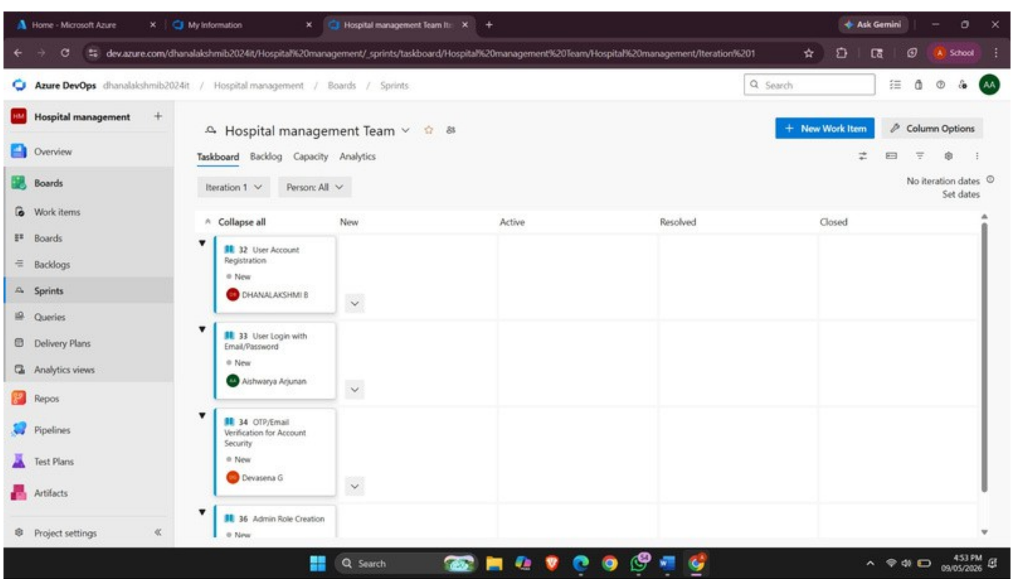
To assign user stories to specific sprints for the Todo list app to setup remainders project.

Sprint Planning

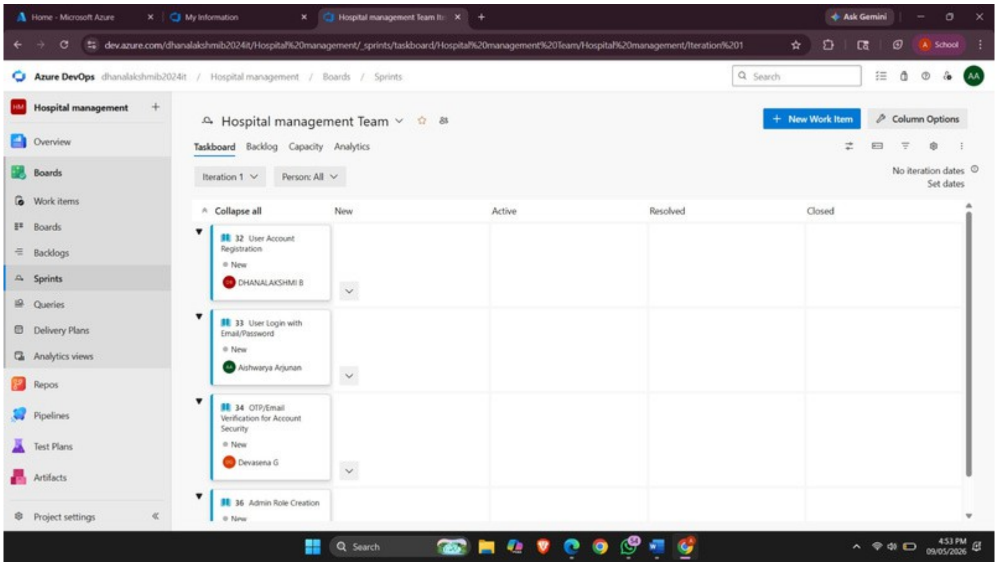
Sprint 1



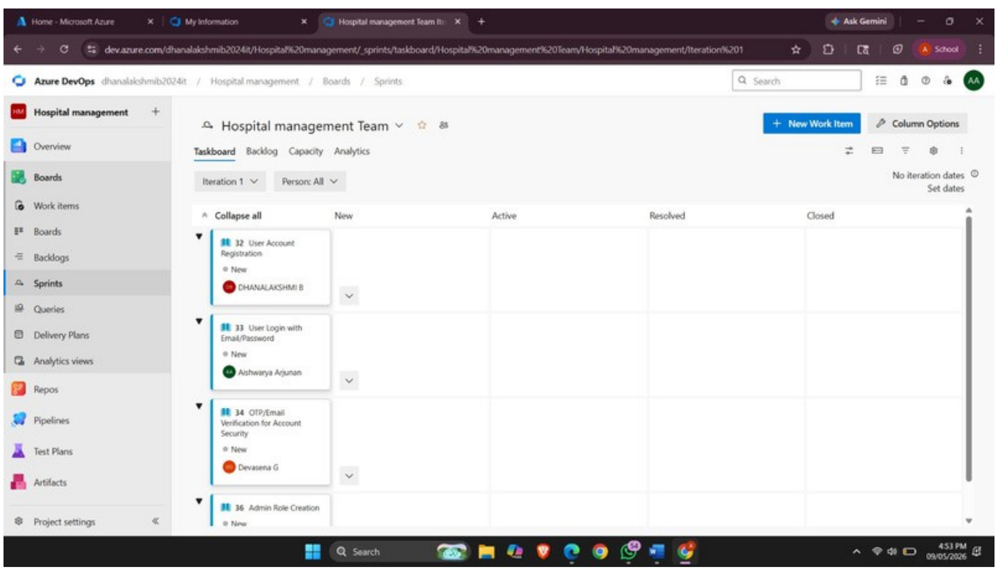
Sprint 2



Sprint 3



Sprint 4



RESULT

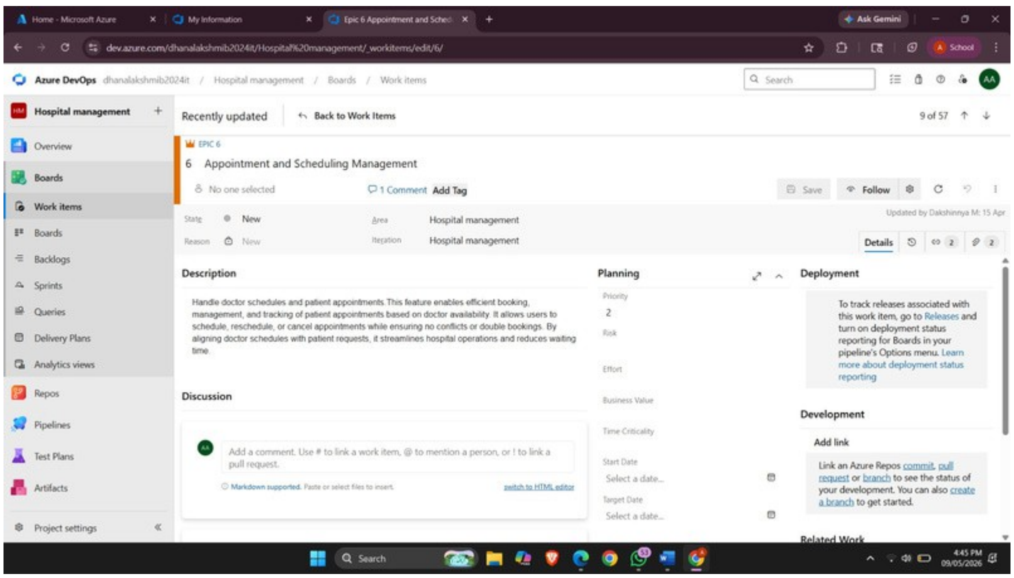
The sprints are created for the todo list app project

EXP NO: 5	**POKER ESTIMATION**

Aim:

Create Poker Estimation for the user stories of the Hospital Management System project.

Poker Estimation

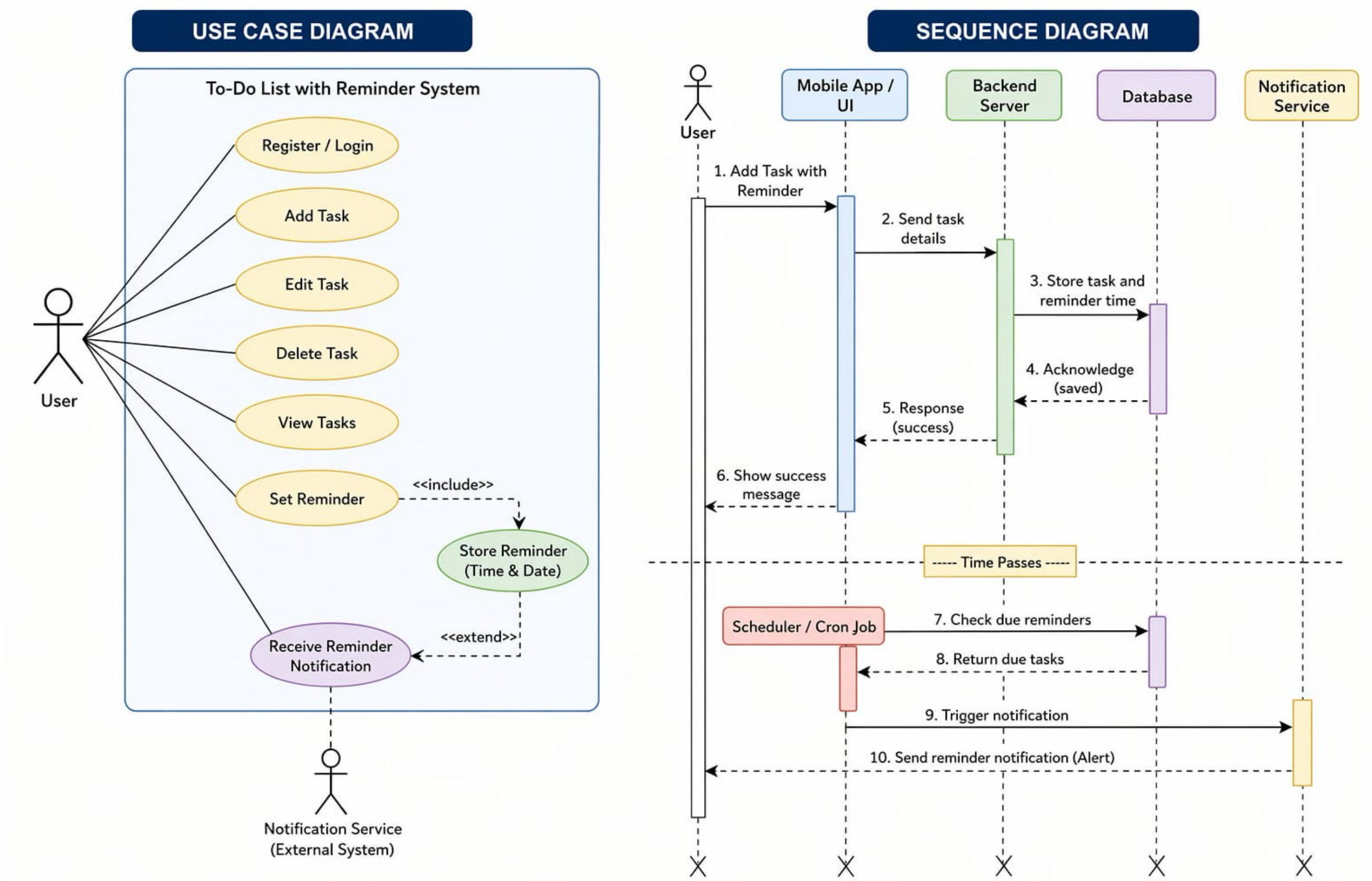


Result

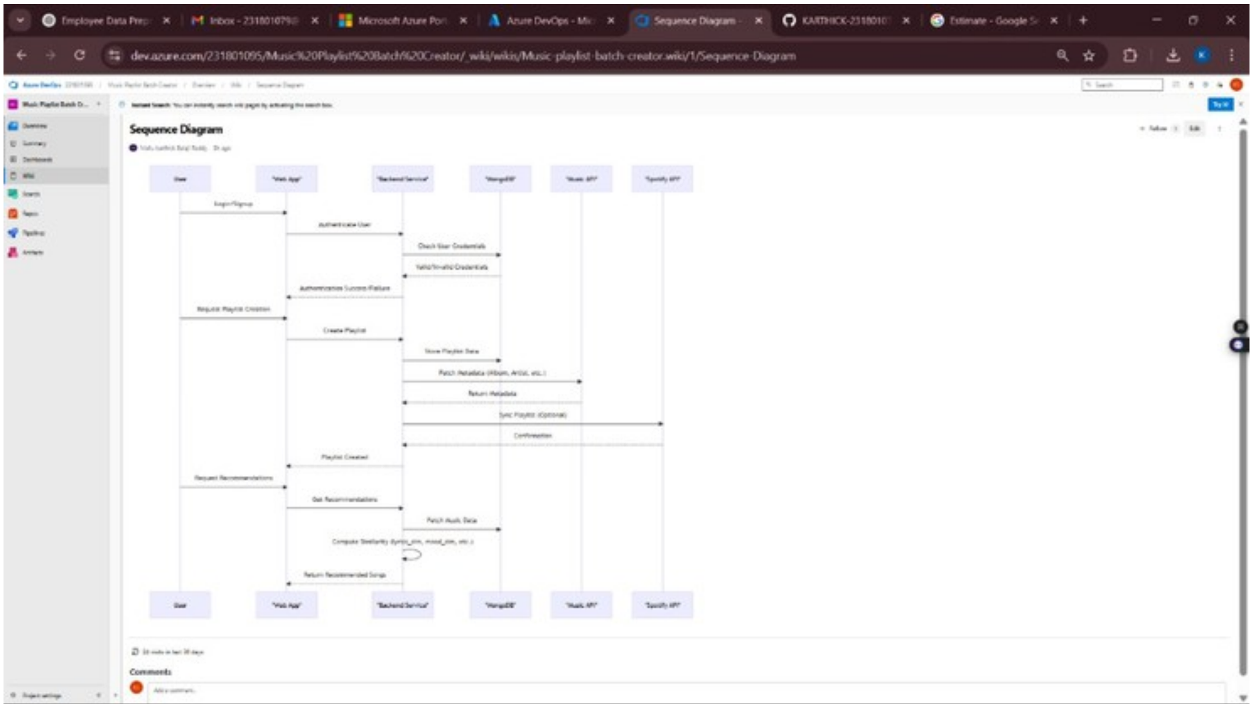
The Estimation/Story Points is created for the project using Poker Estimation.

Ex.No 6	**DESIGNINGCLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE**

Aim: To Design a Class Diagram and Sequence Diagram for the given Project.



6A. Sequence Diagram



Result:

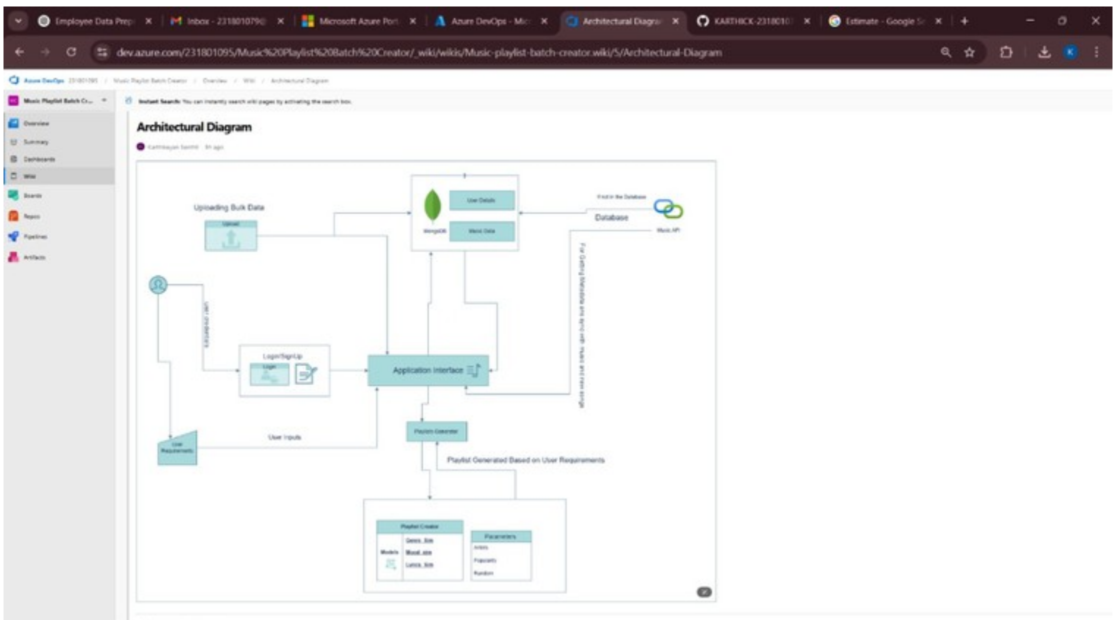
The Class Diagram and Sequence Diagram is designed Successfully for the todo list app Management System.

EX.NO 7	**DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE**

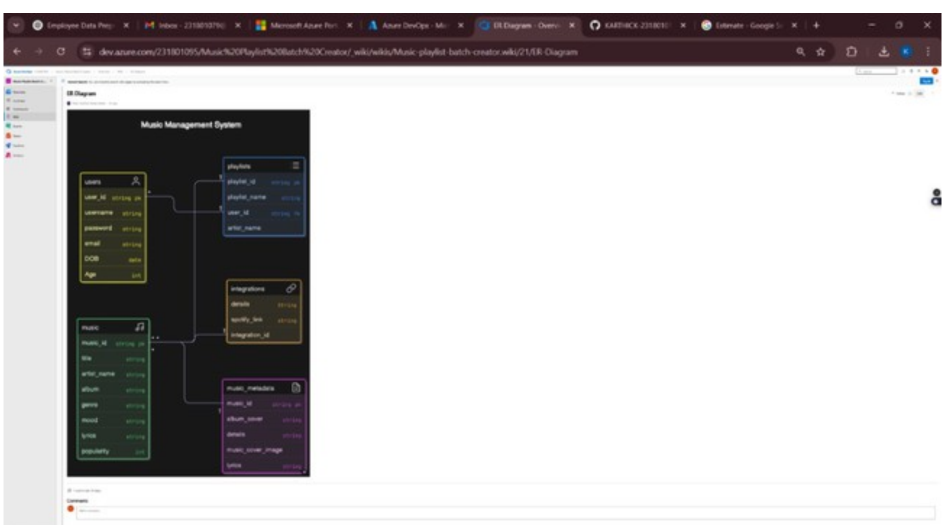
Aim:

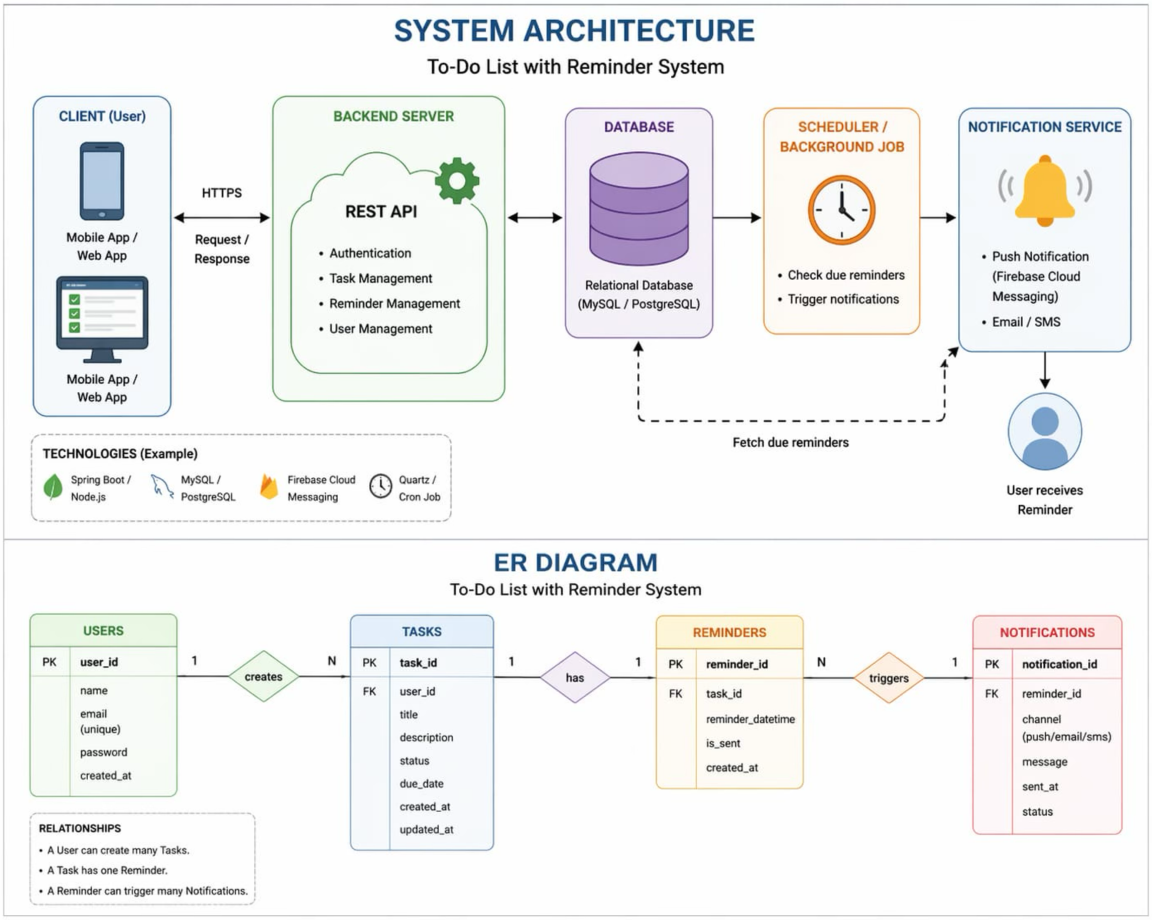
To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram





Result:

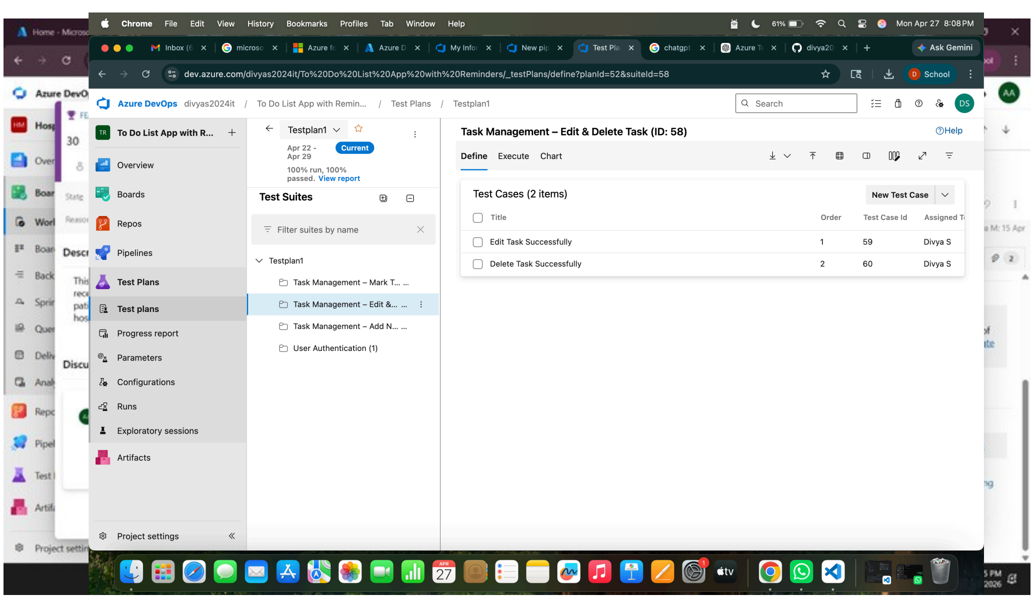
The Architecture Diagram and ER Diagram is designed Successfully for the todo list management system

EXP NO: 8	TESTING - TEST PLANS AND TEST CASES

Aim:

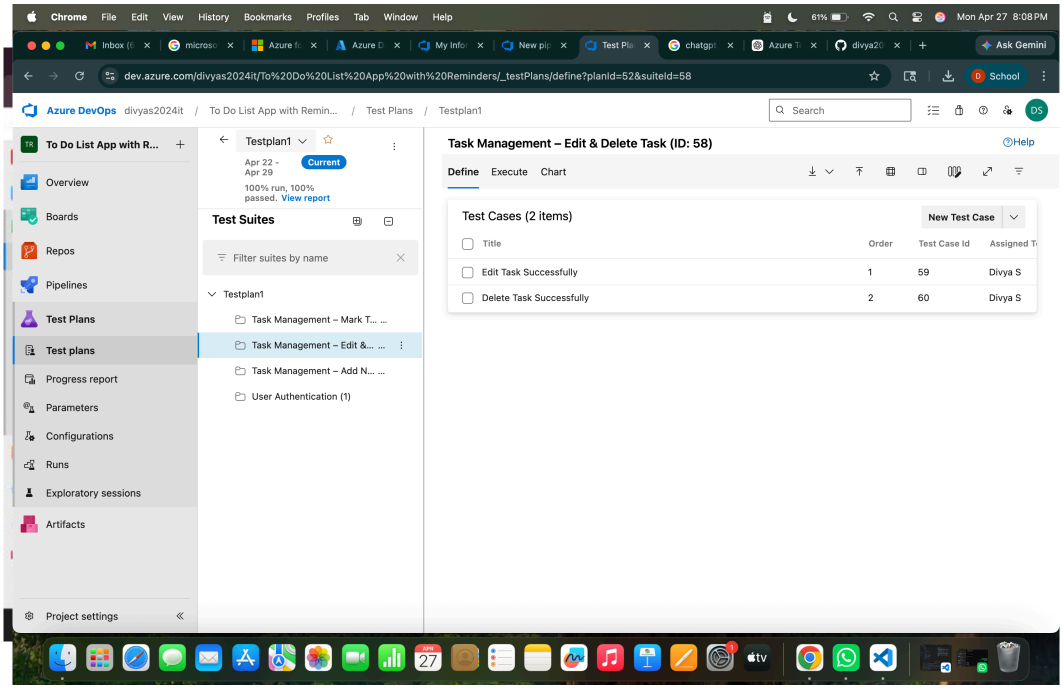
Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

1.New test plan



2.

3.Testsuite



Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Hospital Management System – Test Plans

USER STORIES

- As a patient, I want to register and log in securely so that I can access my medical records (ID: 101).
- As a patient, I want to book a doctor appointment so that I can schedule my consultation (ID:
- 102).
- As a billing officer, I want to generate invoices automatically so that billing is accurate (ID: 103). As a lab technician, I want to upload test results so that doctors can access them instantly (ID:
- 104).

As a doctor, I want to view my appointment schedule so that I can plan my day (ID: 105).

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. **TC01 - Successful Patient Registration**
   - o **Action:**
     - **Test case**
       - Go to the Registration page.
       - Enter valid name, email, date of birth, gender, and password.
       - Click "Register".
   - o **Expected Results:**
       - Registration form is displayed.
       - All fields accept valid input without errors.
       - Patient account is created and user is redirected to the patient dashboard.

- **Type**: Happy Path

2. **TC02 – Secure Login**
    - **Action:**
        - Go to the Login page.
        - Enter valid email and password.
        - Click on "Login".
    - **Expected Results:**
        - Login form is displayed.
        - Fields accept data without error.
        - User is logged in and redirected to the dashboard.
    - **Type:** Happy Path

3. **TC03 – Registration with Duplicate Email**

   - **Action:**
     - Go to the Registration page.
     - Enter a name and an already registered email address.
     - Click on "Register".
   - **Expected Results:**
     - Fields accept data.
     - Error message "Email already registered" is displayed.
   - **Type:** Error Path

4. **TC04 – Login with Wrong Password**

   - **Action:**
     - Go to the Login page.
     - Enter valid email and incorrect password.
     - Click on "Login".
   - **Expected Results:**
     - Input is accepted.
     - Error message "Invalid username or password" is shown.
   - **Type:** Error Path

Test Suit: TS02 - Appointment Booking (ID: 111)

1. **TC05 – Book Appointment Successfully**

   - **Action:**
     - Log in successfully.
     - Search for a doctor by specialization and select an available time slot. Click "Book Appointment".
   - **Expected Results:**
     - Appointment is confirmed and a confirmation notification is sent to the patient.
   - **Type:** Happy Path

2. TC06 – Book Appointment for Unavailable Slot

   - **Action:**
     - Select a doctor with no available slots.
     - Try to book an appointment.
   - **Expected Results:**
     - Doctor has no available slots for the selected date.
     - Error message "No available slots for the selected doctor" is shown.
   - **Type:** Error Path

Test Suit: TS03 - Billing & Invoice (ID: 112)

1. **TC07 – Generate Invoice Successfully**
   - **Action:**

- Log in as billing officer. Select a completed appointment. Click "Generate Invoice".
- Verify the generated invoice details.

- **Expected Results:**
  - Invoice is generated with correct service charges, patient details, and total amount.
- **Type:** Happy Path

2. TC08 – Invoice Generation Fails Offline

   - **Action:**
     - Disconnect network. Attempt to generate invoice.
     - Check the error response on the billing screen.
   - **Expected Results:**
     - Error message "Unable to generate invoice. Check your connection" is shown.
   - **Type:** Error Path

Test Suit: TS04 - Lab Test Results (ID: 113)

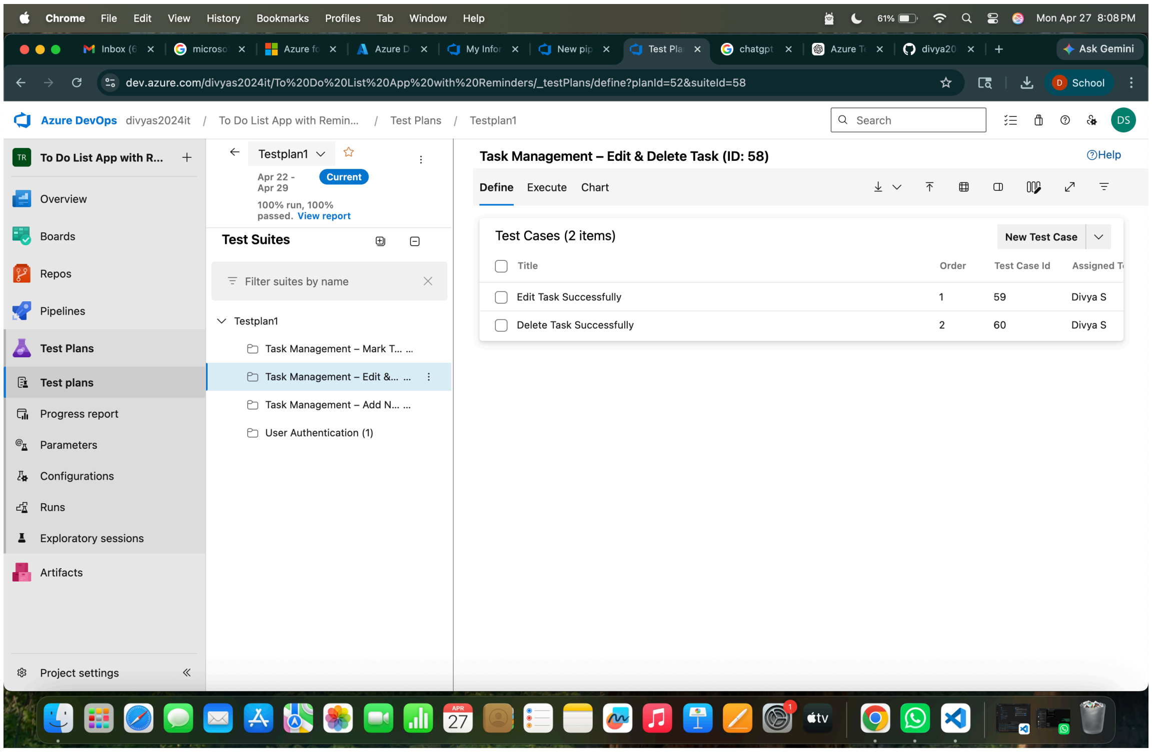
1. **TC09 - Upload Lab Test Result Successfully**
   - **Action:**
     - Log in as lab technician. Select a patient test request.
     - Upload result file and click "Submit".
     - Verify the upload confirmation.
   - **Expected Results:**
     - Result is uploaded successfully and the doctor is notified.
   - **Type:** Happy Path
2. **TC10 - Upload Fails with Unsupported File Format**
   - **Action:**
     - Attempt to upload a non-PDF/image file (e.g., .exe) as a lab result.
     - Click "Submit" without selecting a valid file.
     - Check the error message on the upload form.
   - **Expected Results:**
     - Error message "Invalid file format. Only PDF and image files are accepted" is shown.
   - **Type:** Error Path
3. **TC11 - Doctor Views Appointment Schedule**
   - **Action:**
     - Log in as doctor. Navigate to "My Schedule".
     - View today's appointments.
     - Verify appointment details are displayed correctly.
   - **Expected Results:**
     - All scheduled appointments for the day are displayed with patient details.
   - **Type:** Happy Path
4. **TC12 - Schedule Fails to Load**

- **Action:**
  - Log in as doctor. Simulate a server error.
  - Navigate to "My Schedule".
  - Check the error response on the schedule page.
  - Attempt to load the schedule.
- **Expected Results:**
  - Error message "Unable to load schedule. Please try again later" is displayed.
- **Type:** Error Path

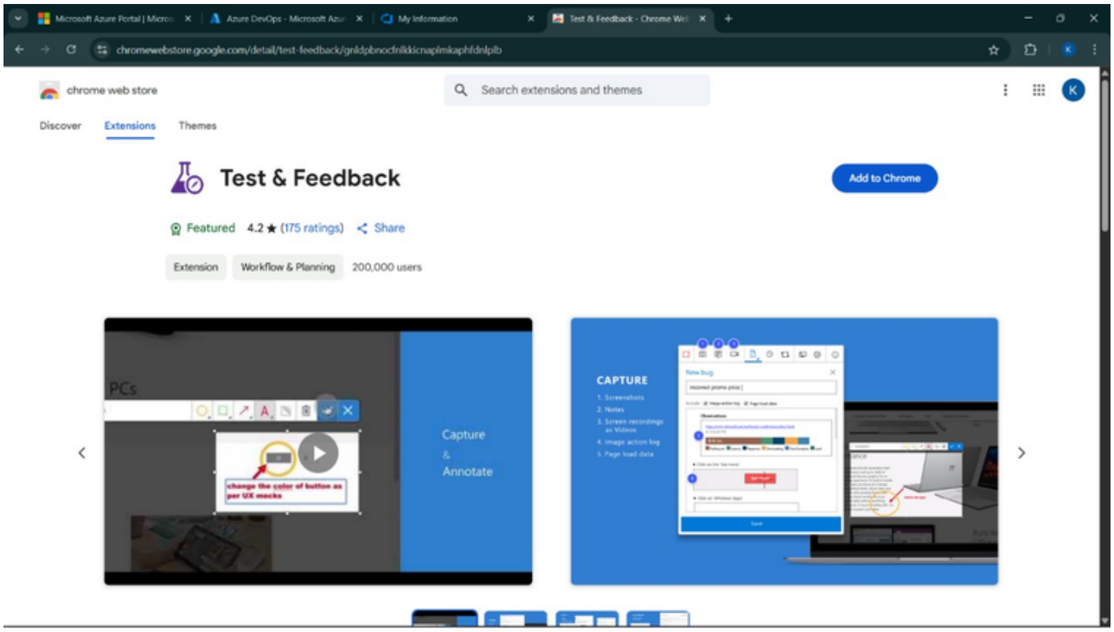
Test Suit: TS05 - Pharmacy & Prescription (ID: 114)

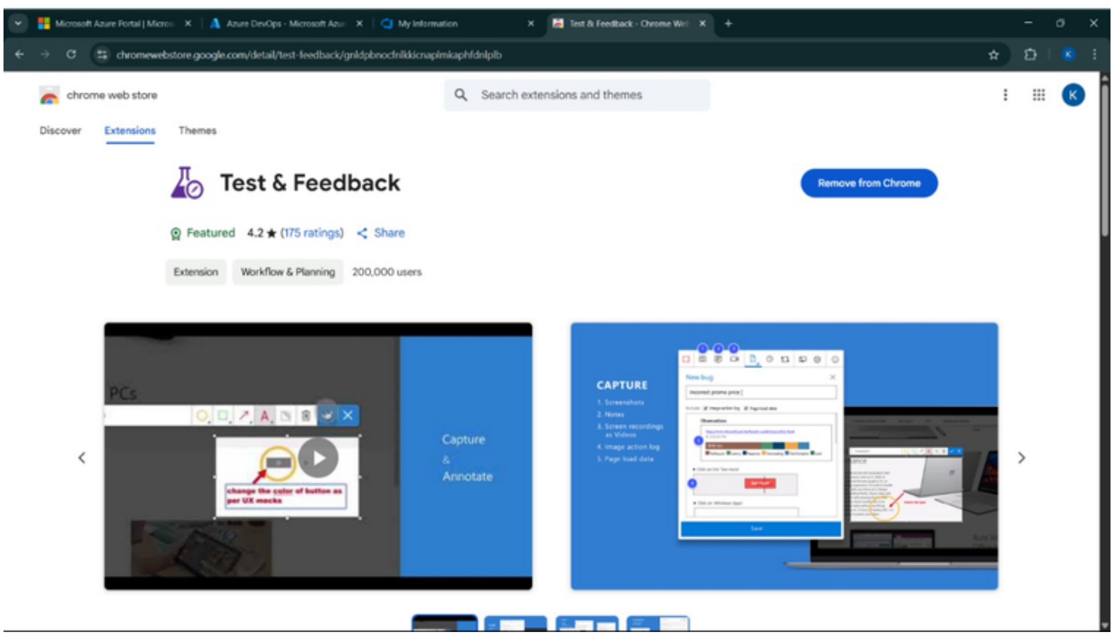
1. **TC13 – Add Prescription Successfully**
   - **Action:**
     - Log in as doctor. Open a patient record.
     - Click "Add Prescription". Enter medicine name, dosage, and duration.
     - Click "Save Prescription".
     - Verify the prescription is saved.
   - **Expected Results:**
     - Prescription is saved successfully and is visible in the patient's record.
   - **Type:** Happy Path
2. **TC14 – Add Prescription Fails with Empty Fields**
   - **Action:**
     - Log in as doctor. Open a patient record.
     - Click "Add Prescription" and leave medicine name blank.
     - Click "Save Prescription" without filling required fields.
     - Check the error response on the prescription form.
   - **Expected Results:**
     - Error message: "Please fill in all required fields" is displayed.
   - **Type:** Error Path

Test Cases



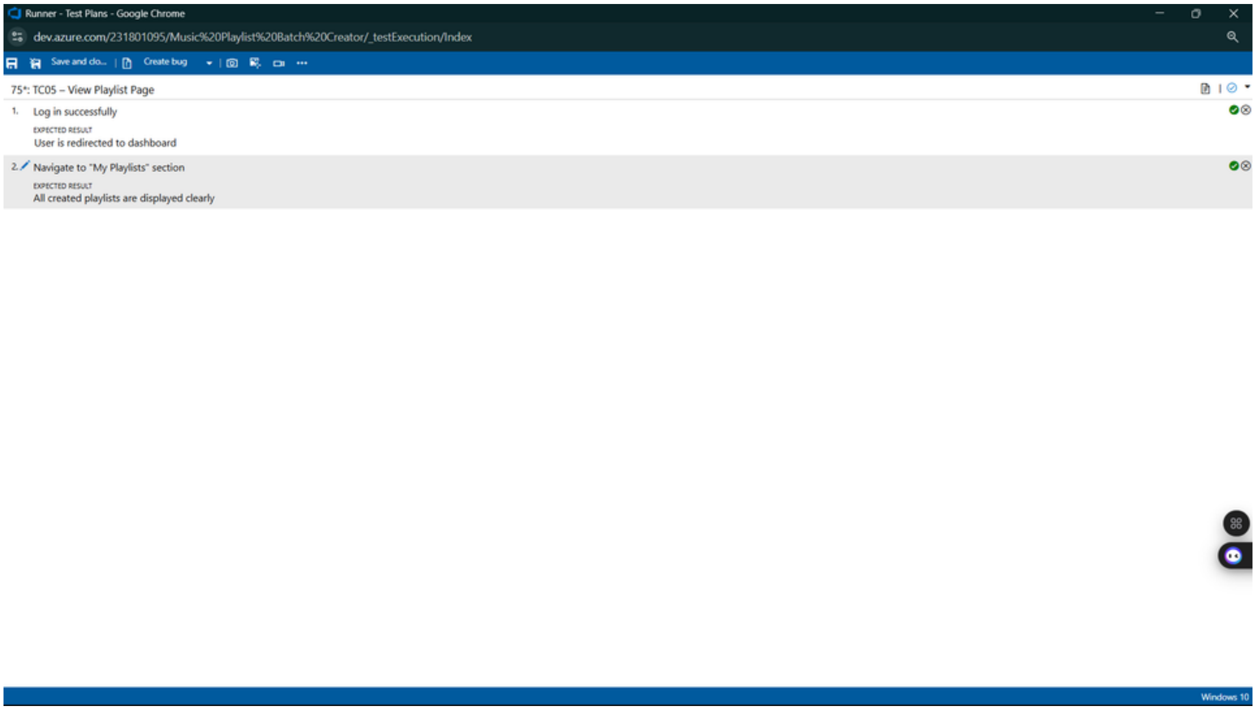
4.Installation of test



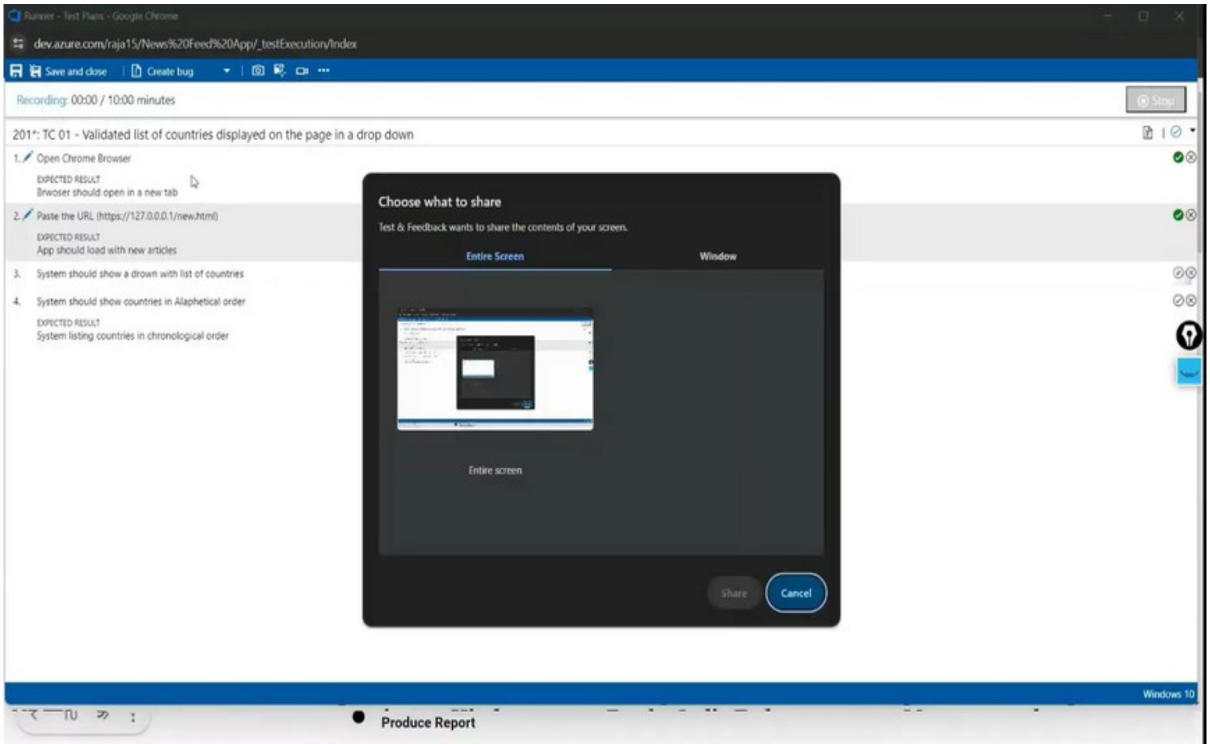


Test and feedback
Showing it as an
extension

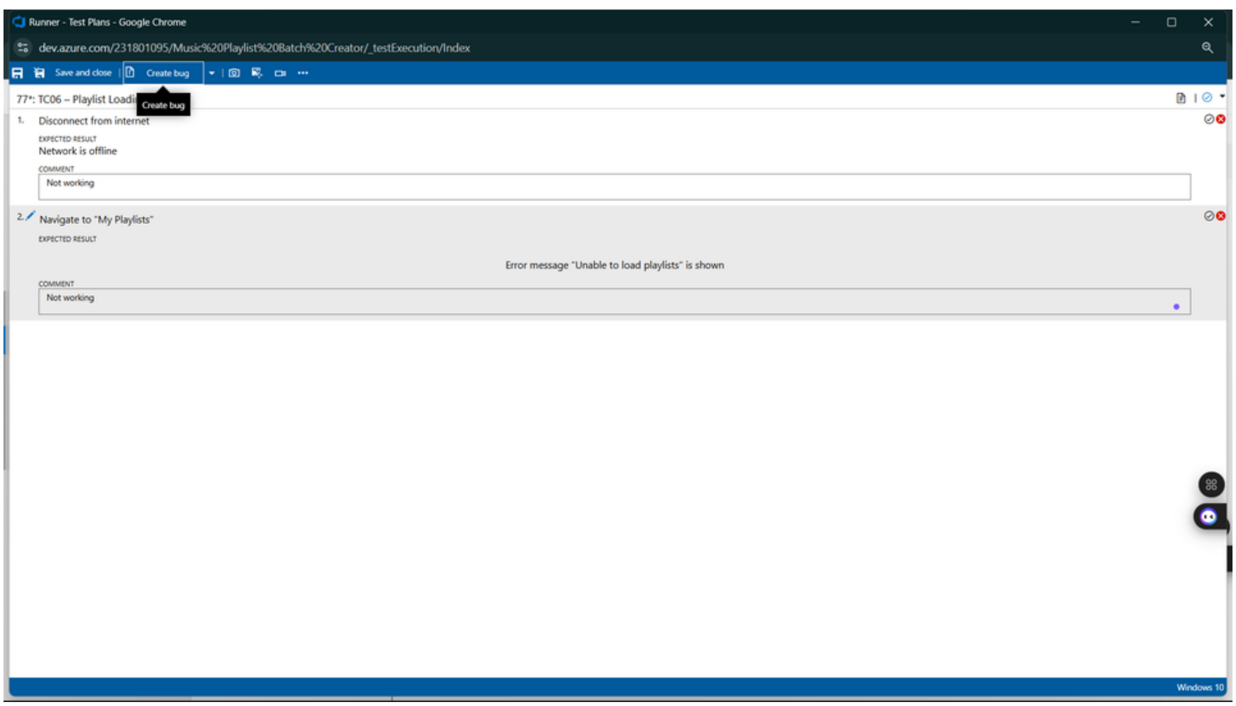
5.Running the test cases

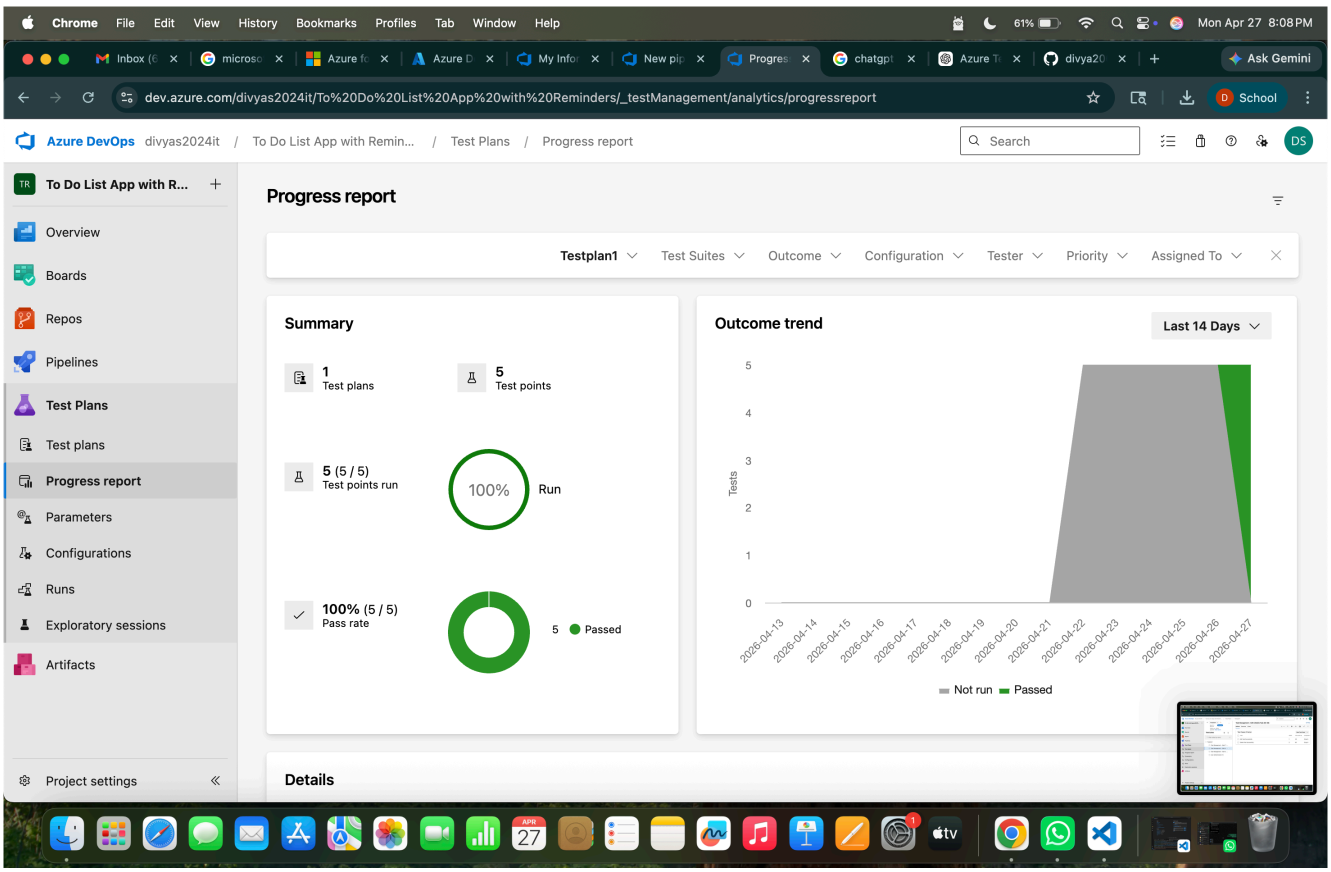


6.Recording the test case

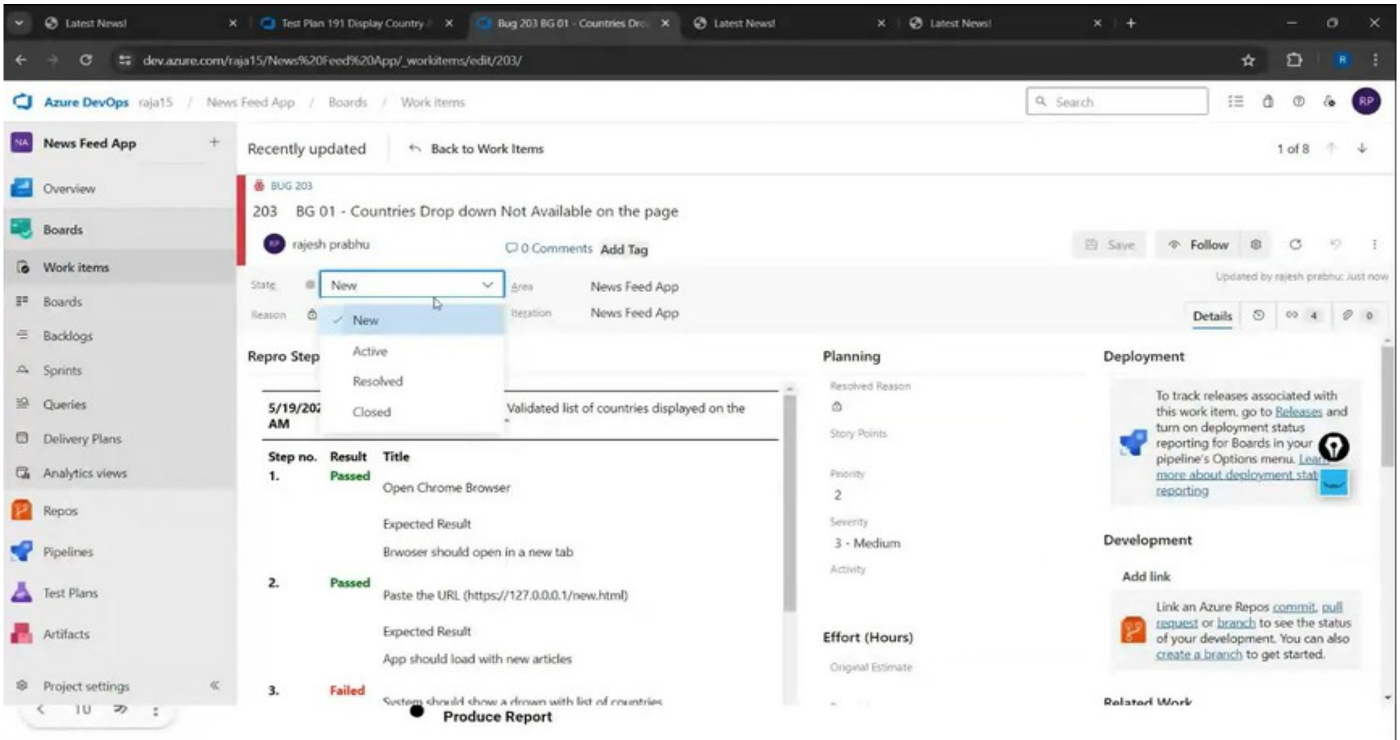


7.Creating the bug

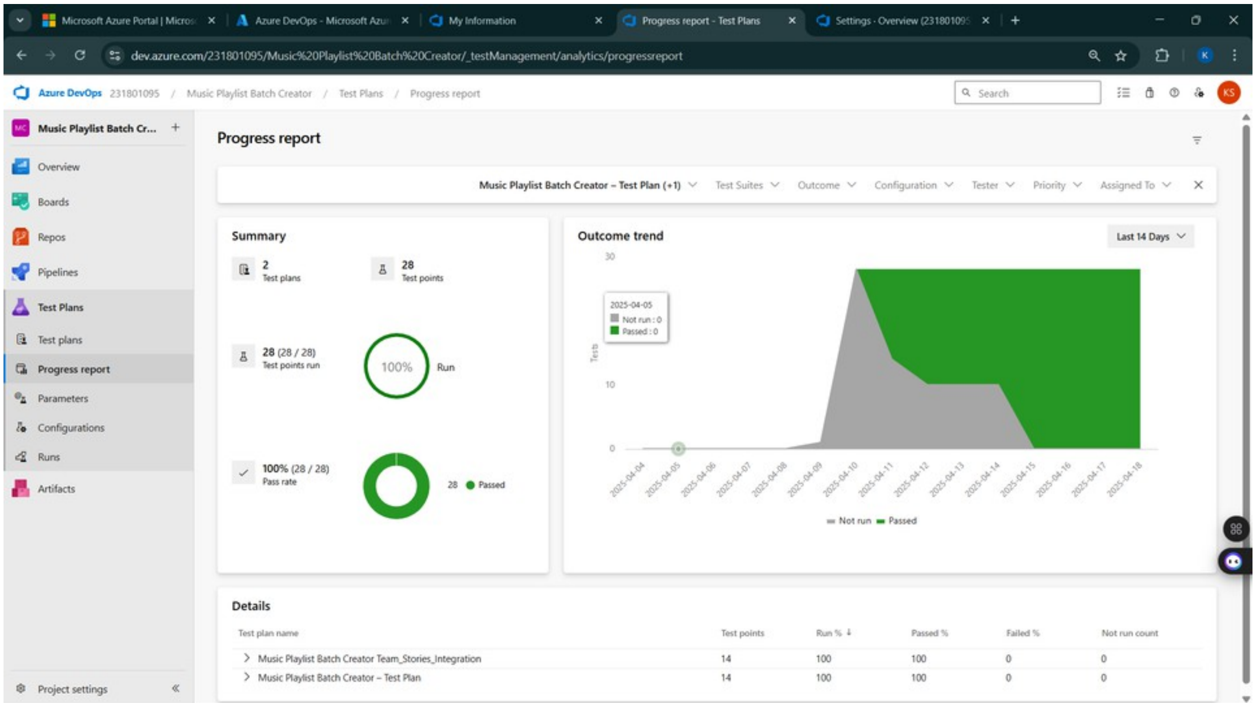




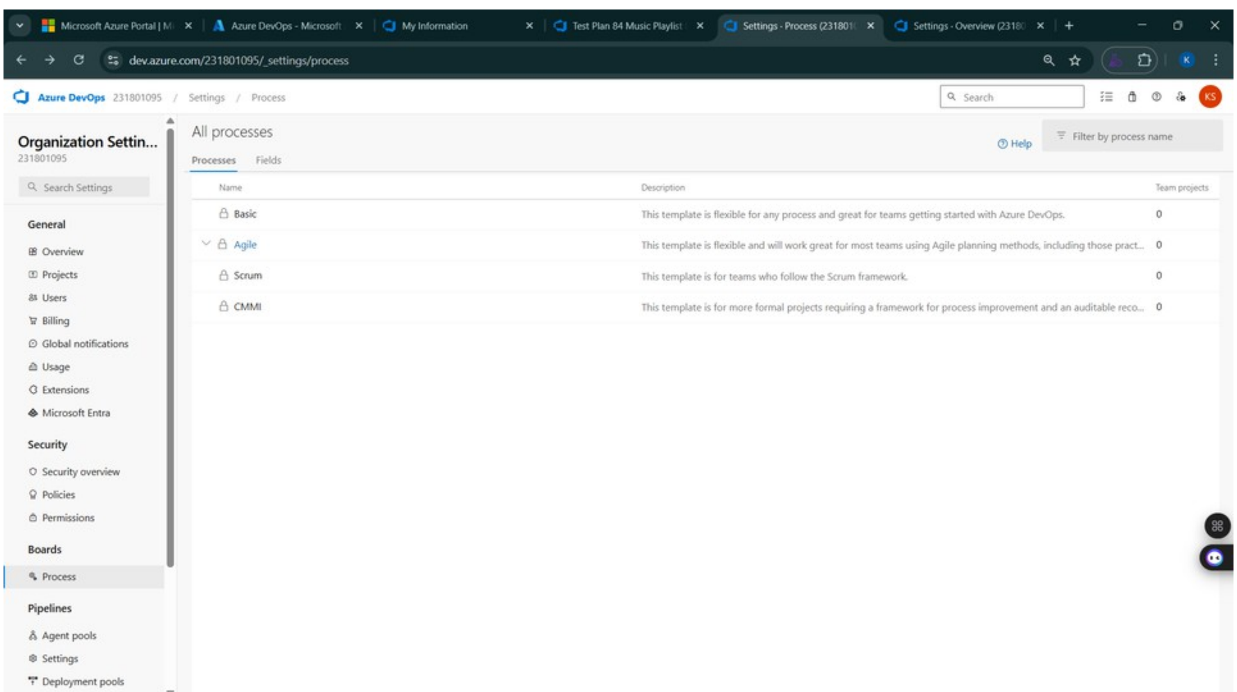
8.Test case results

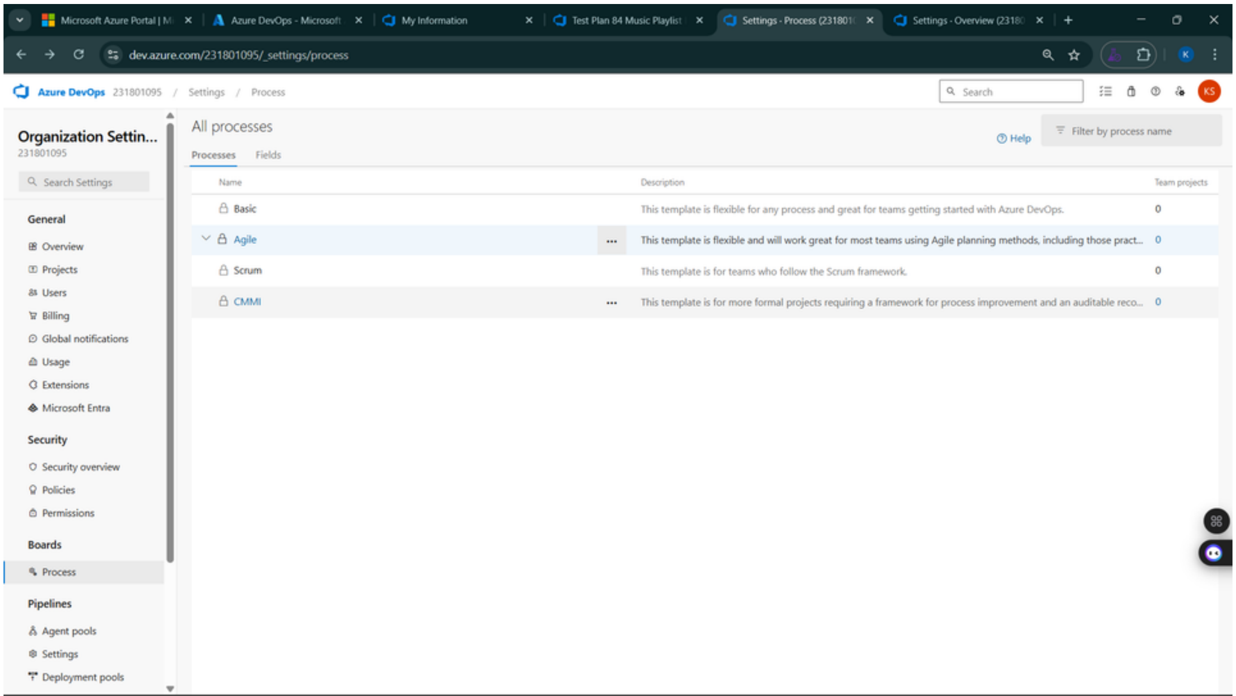


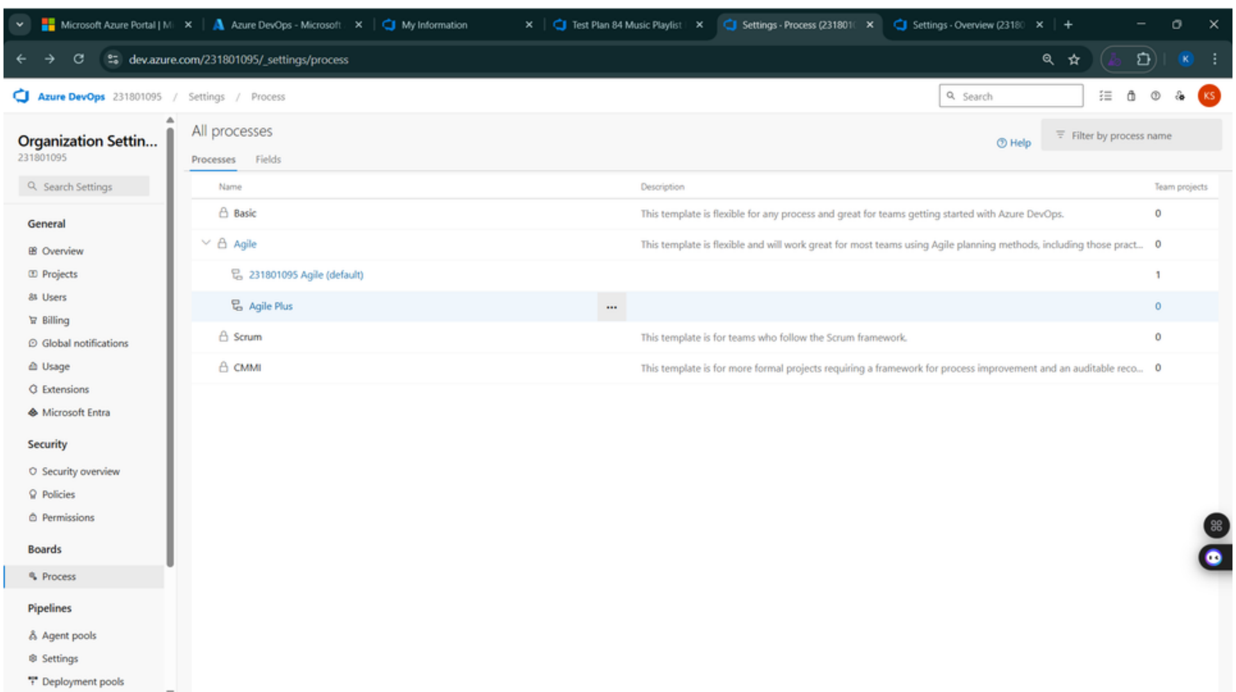
9. Test report summary



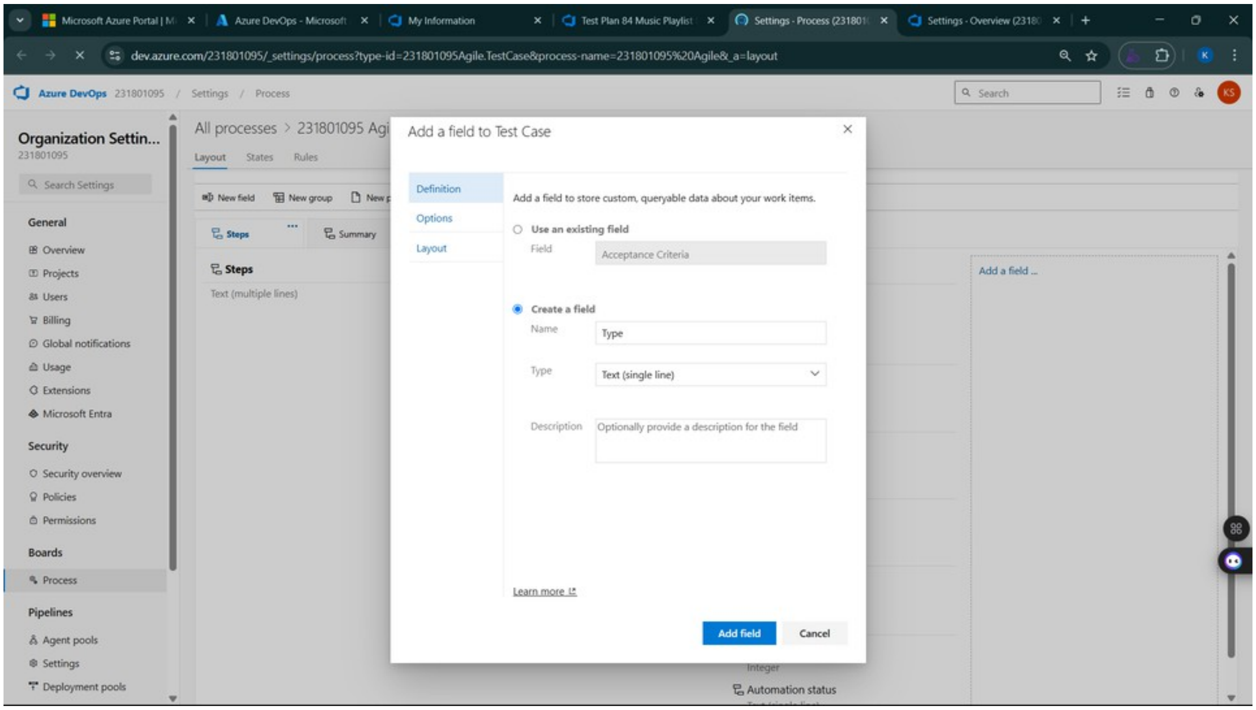
11. Changing the test template

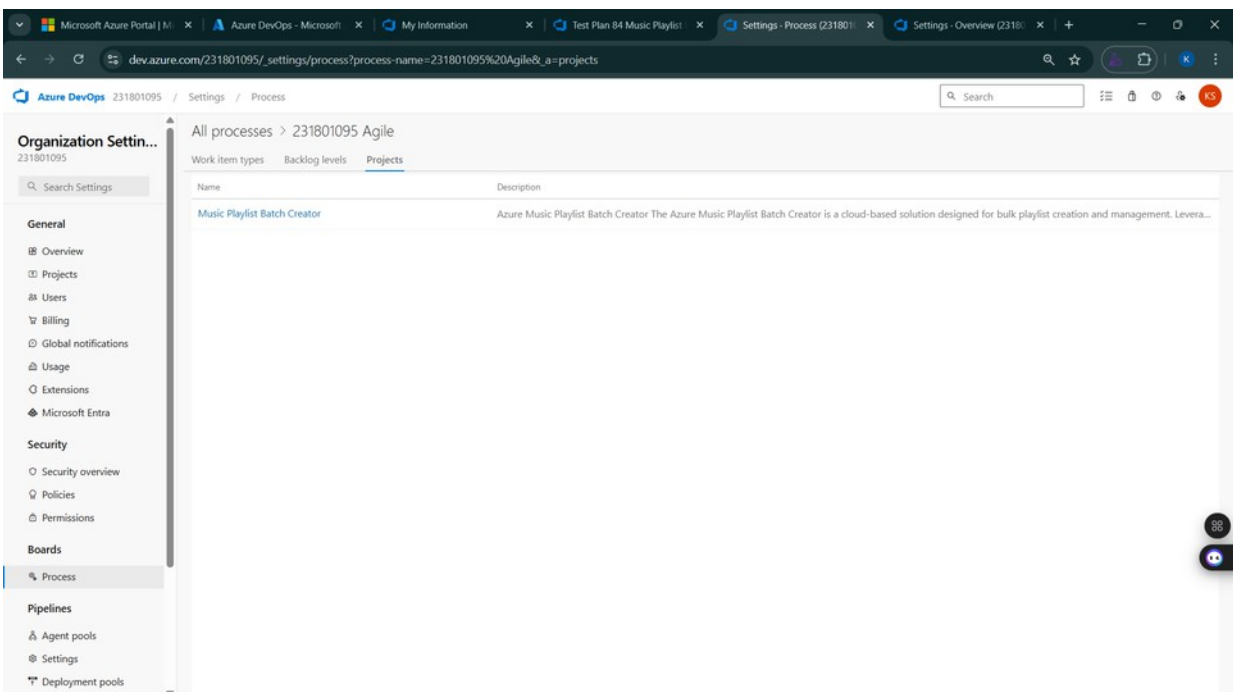


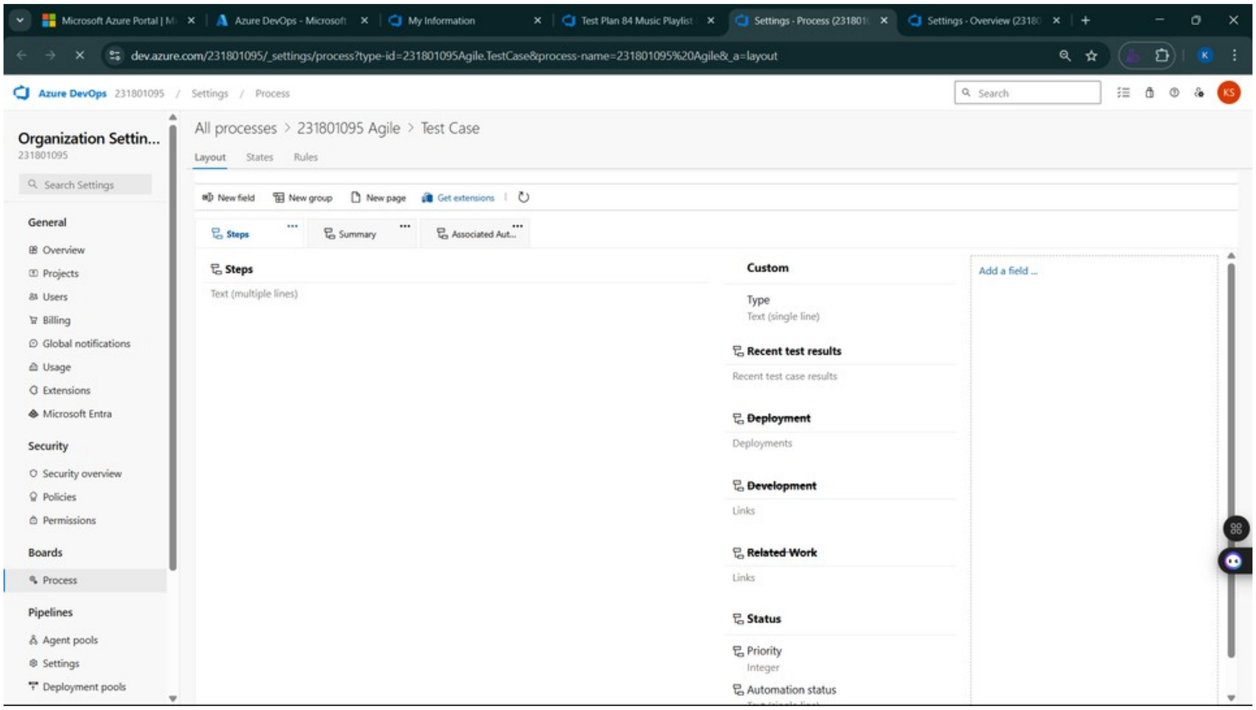

dev.azure.com/231801095/_settings/process
Azure DevOps 231801095 / Settings / Process
Organization Settin...
All processes
Processes Fields
Basic
Agile
Scrum
CMMI
This template is flexible for any process and great for teams getting started with Azure DevOps.
This template is for teams who follow the Scrum framework.


dev.azure.com/231801095/_settings/process
Azure DevOps 231801095 / Settings / Process
Organization Settin...
All processes
Processes Fields
Basic
Agile
231801095 Agile (default)
Agile Plus
Scrum
CMMI

12. View the new test case template







Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9	LOAD TESTING AND PIPELINES

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
   Go to https://portal.azure.com and log in.
· Create the Resource
   - Go to *Create a resource* → Search for "Azure Load Testing".
2. - Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
·
   - Subscription*n:* Choose your Azure subscription.
   - *Resource Group:* Create new or select an existing one.
   - *Name:* Provide a unique name (no special characters).
   - *Location:* Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
. Click Review + Create, then Create.
5. Once deployment is complete, click Go to resource.
. 6

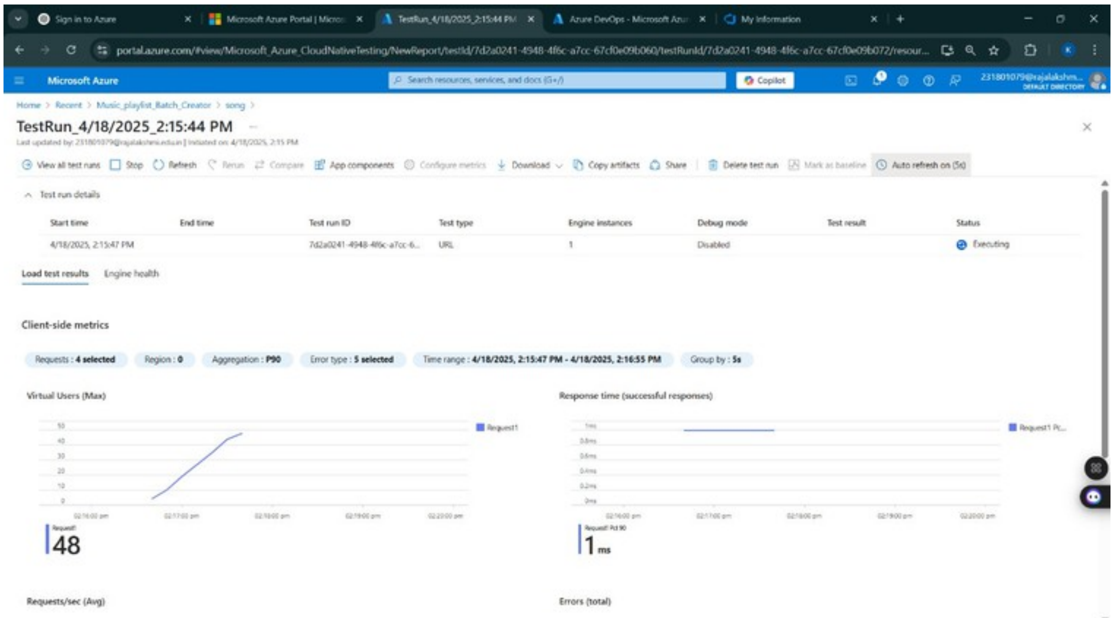
Steps to Create and Run a Load Test:

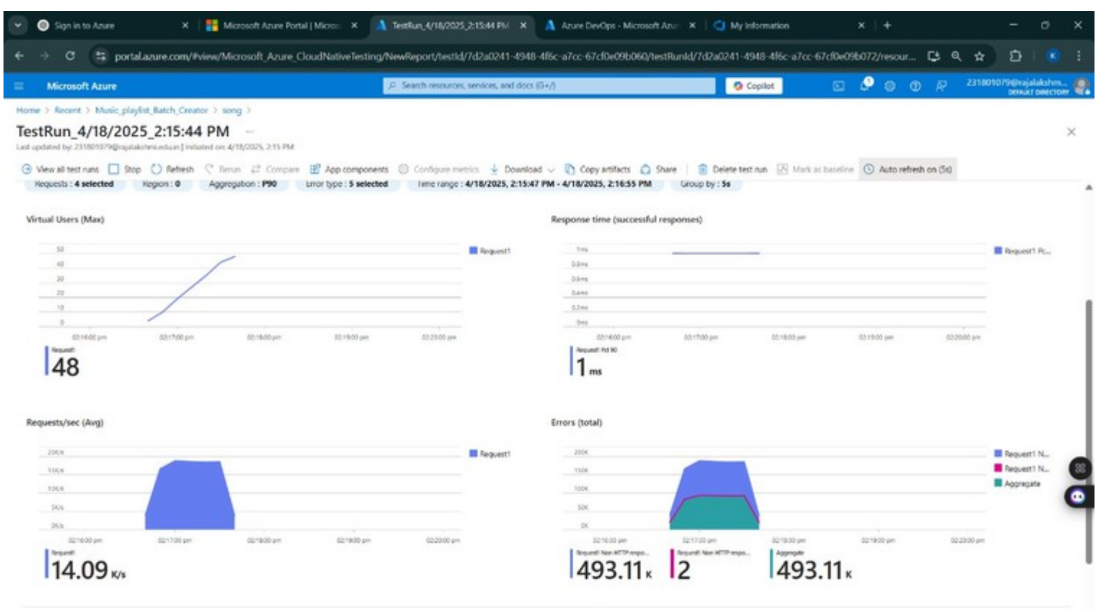
Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.
. Basics Tab
2. - *Test Name:* Provide a unique name.
·
   - *Description:* (Optional) Add test purpose.
   - *Run After Creation:* Keep checked.
3. Load Settings
   - *Test URL:* Enter the target endpoint (e.g., https://yourapi.com/products).

4. Click Review + Create → Create to start the test.

Load Testing





Pipelines

Description:
This experiment demonstrates how to connect a GitHub-hosted Node.js/Express-based Hospital Management System project with Azure DevOps. The pipeline will automatically install dependencies, run tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
   - In Azure DevOps, create a new project.
   - Create a pipeline and select GitHub as the source.
   - Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
   - In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
   - Add the following basic pipeline configuration for Node.js and Express:

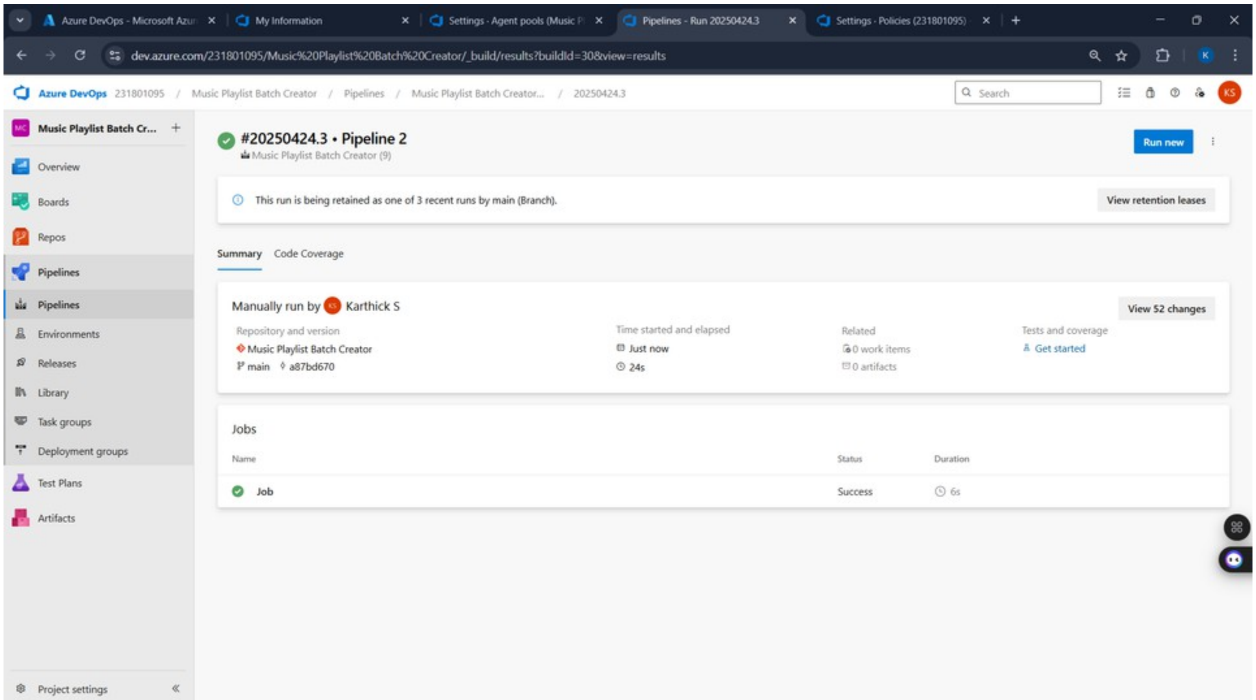
yml Code

```
trigger:
 - main # Trigger pipeline when changes are pushed to the main branch

pool:
 vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:
# Step 1: Checkout the code from GitHub
- checkout: self

# Step 2: Set up Node.js environment
- task:
 NodeTool@0
 inputs:
   versionSpec: '18.x' # Use the latest Node.js 18.x version
 displayName: "Set up Python"

# Step 3: Install dependencies from the correct path
- script: |
 python -m pip install --upgrade pip
 npm install # Install HMS project dependencies
 displayName: "Install dependencies"

# Step 4: Run a simple Python script to check the environment
- script: |
 node -e "console.log(' Hospital Management System pipeline running
successfully!')" displayName: "Run a Python script"
```

3. Pipeline Tasks Include:
   - Setting up the Node.js environment using the NodeTool task.
   - Installing project dependencies using npm install from package.json.
   - Running unit tests using npm test to verify all HMS modules work correctly.

4. Run and Monitor Pipeline:
   - Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
   - Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline



Result:

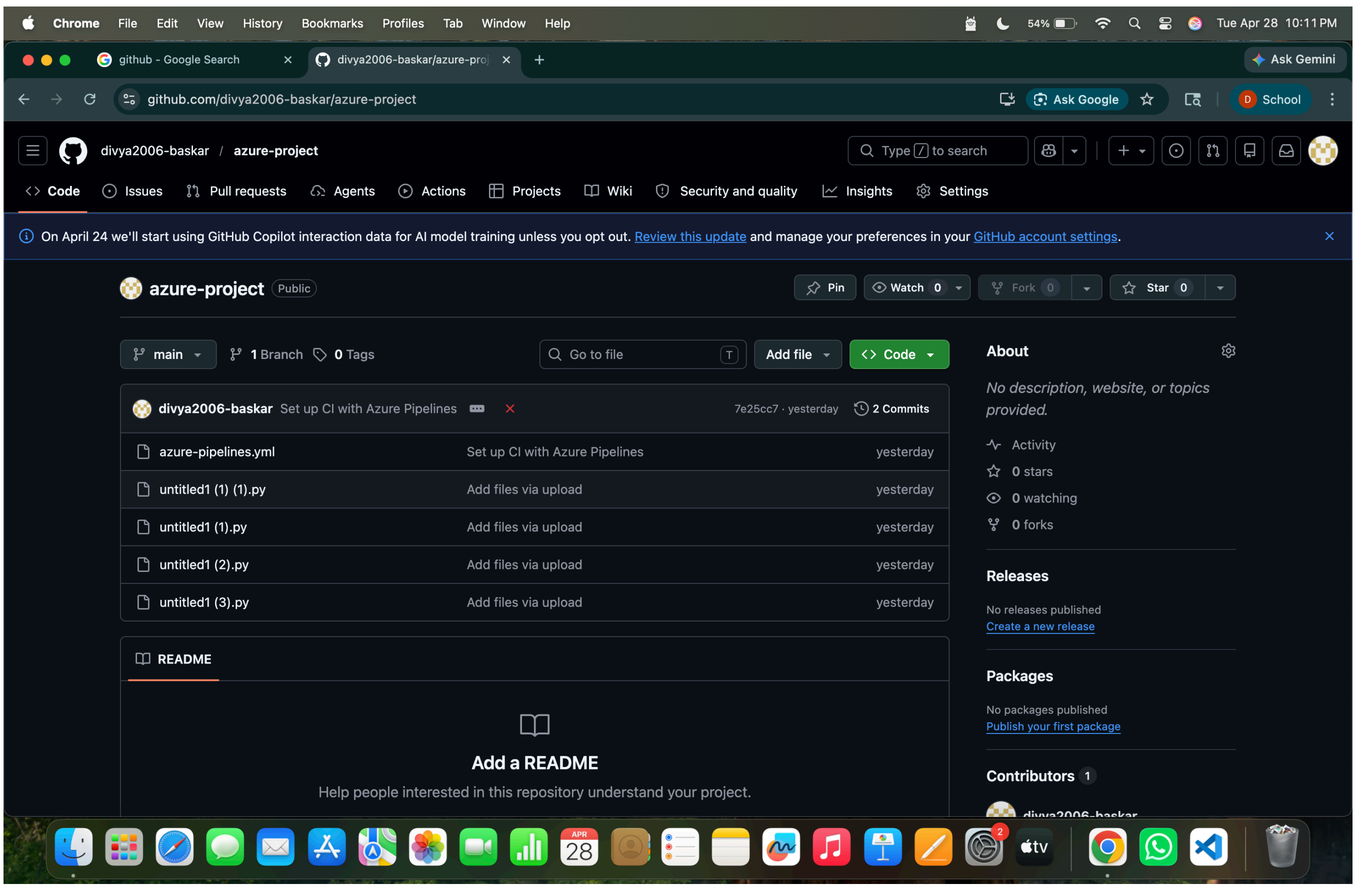
Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

EXP NO: 10	GITHUB:PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the Todo List Management System folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the HMS project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.